prithvika

a coal land reclamation, research and innovation campus

thesis - 2026

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LITERATURE STUDY

INTRODUCTION

The design of a contemporary research-based institutional campus requires a clear understanding of functional activities, spatial standards, safety norms, and infrastructural requirements. This literature study examines the essential spatial and functional parameters for a campus comprising research facilities, academic spaces, exhibition areas, and tourism-oriented public interfaces.

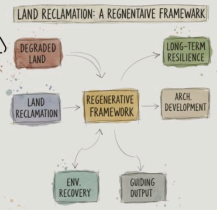
The extracted literature data informs the area program, zoning strategy, circulation hierarchy, and service planning of the proposed thesis project, ensuring that the design decisions are technically sound, code-compliant, and functionally efficient.

SCOPE OF STUDY

- Research facilities
- Academic spaces
- Exhibition & public outreach
- Tourism & visitor facilities

LAND RECLAMATION

Land reclamation is the process of restoring degraded or disturbed land to a stable and usable condition through ecological and spatial interventions. It focuses on improving soil health, landform, and water systems to support future use.

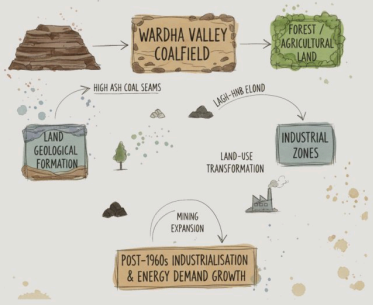


In this thesis, land reclamation is viewed as a regenerative framework where environmental recovery and architectural development occur together. The reclaimed land acts as a base for informed planning, guiding built form, zoning, and landscape design while ensuring long-term ecological resilience and usability.

HISTORY & BACKGROUND STUDY

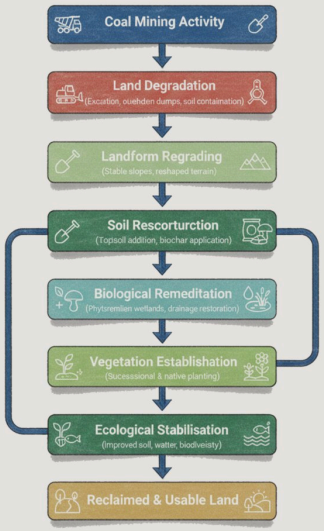
Chandrapur Coal Mining

- Chandrapur district forms a core part of the **Wardha Valley Coalfield.**
- Coal seams belong to the **Gondwana geological formation**, known for high ash content coal.
- Coal availability **directly influenced land-use** transformation from forest/agricultural land to industrial zones
- Mining expansion **intensified after post-1960s industrialisation** and energy demand growth



COAL LAND RECLAMATION

Coal land reclamation refers to the planned process of restoring land disturbed by coal mining to a stable, environmentally functional, and usable condition. Mining activities significantly alter landforms, soil structure, hydrological systems, and ecological balance, necessitating systematic intervention after extraction. Reclamation therefore aims to re-establish physical stability and ecological processes while enabling future land use compatible with environmental constraints.



Coal land reclamation is increasingly understood as a regenerative practice rather than mere land repair. It integrates landform correction, soil reconstruction, water management, and biological remediation to transform degraded mining landscapes into resilient systems capable of supporting ecological, social, and developmental functions.

REGENERATION PROCESSES

- **Phytoremediation (Heavy Metals)**

Phytoremediation uses hyperaccumulator plant species to absorb and immobilize heavy metals from contaminated soils. This process gradually reduces soil toxicity while allowing surface land use.

- **Mycoremediation**

Mycoremediation employs fungi to degrade and transform toxic contaminants through enzymatic action. It is effective in mining landscapes where organic and industrial pollutants are present.

- **Biochar Application**

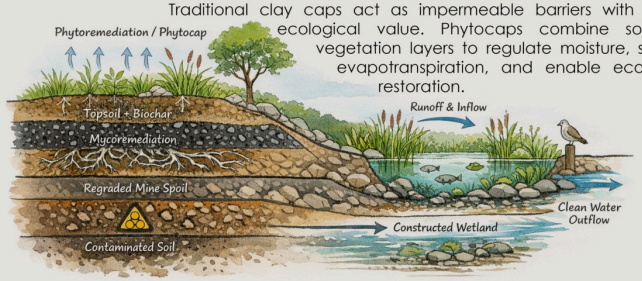
Biochar is incorporated into degraded soil to improve fertility, moisture retention, and microbial activity. It also enables long-term carbon sequestration within reclaimed land.

- **Constructed Ecosystems (Wetlands)**

Constructed wetlands use vegetation, soil, and microbial processes to treat polluted mine runoff. These systems simultaneously improve water quality and create ecological habitats.

- **Alternative Capping (Phytocaps vs. Clay Caps)**

Traditional clay caps act as impermeable barriers with limited ecological value. Phytocaps combine soil and vegetation layers to regulate moisture, support evapotranspiration, and enable ecological restoration.



CHALLENGES WITH POST-MINED LAND

- **Unstable Landform**

Excavation pits and overburden dumps create uneven terrain with slope instability and subsidence risks.

- **Poor Soil Quality**

Loss of topsoil, compaction, and contamination reduce fertility and load-bearing capacity.

- **Altered Hydrology**

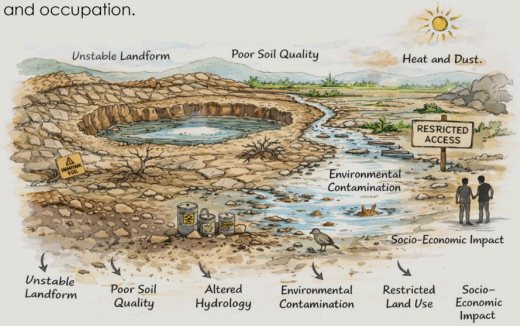
Disrupted drainage patterns cause waterlogging, erosion, and reduced groundwater recharge.

- **Environmental Contamination**

Presence of heavy metals, coal residue, and industrial pollutants affects soil and water systems.

- **Restricted Land Use**

Safety, pollution, and legal constraints limit immediate construction and occupation.



MATERIAL INNOVATIONS

1. Waste-Based Materials

- **Fly Ash Concrete**

Utilizes coal combustion waste as a partial cement replacement (25–35%), reducing embodied energy by approximately 20%.

- **Mine Tailings in Bricks**

Incorporates up to 40% mine tailings by weight in brick manufacturing, reducing waste disposal and raw material extraction.

2. Low-Impact Construction Assemblies

- **Modular Construction Systems**

Off-site fabrication enables 30–50% faster construction with minimal on-site disturbance. This method is suitable for ecologically sensitive and recently reclaimed land.

- **Helical Pile Foundations**

Installed with nearly 90% less soil disruption compared to conventional driven piles. They are reversible and ideal for weak or remediated soils.

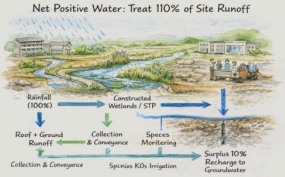
- **Green Roof Systems**

Vegetated roofs retain 50–80% of stormwater runoff and reduce building energy demand by up to 25%. They contribute to microclimate regulation and ecological continuity.

PERFORMANCE METRICS

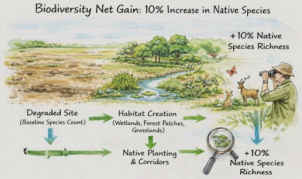
1. Net Positive Water (110% Runoff Treatment)

Net positive water aims to treat and recharge more water than the total runoff generated on site. By managing 110% of runoff through wetlands, recharge zones, and reuse systems, the campus contributes surplus clean water back to the local hydrological system.



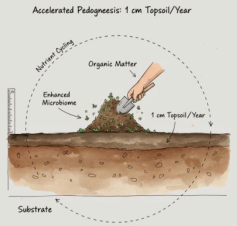
2. Biodiversity Net Gain (10% Increase)

Biodiversity net gain ensures that post-development ecological value exceeds pre-existing conditions. A minimum 10% increase in native species richness is achieved through habitat creation, native planting, and ecological corridors.



3. Soil Creation (1 cm/year)

Accelerated pedogenesis focuses on rebuilding living soil through organic matter, biochar, microbes, and vegetation. Creating approximately 1 cm of topsoil annually indicates active ecological recovery on degraded land.

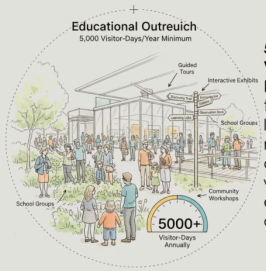


4. Community Employment (25% Local Hiring)

Local hiring integrates surrounding communities into the development process. Ensuring at least 25% employment during construction strengthens socio-economic resilience and local ownership of the project.

5. Educational Outreach (5,000 Visitor-Days/Year)

Educational outreach positions the campus as a public learning interface. Hosting a minimum of 5,000 visitor-days annually through exhibitions, workshops, and guided tours ensures continuous knowledge dissemination.



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LITERATURE STUDY

FUNCTIONAL INTENT

1. Research Facilities (TSS - Industrial → Research Laboratories - Pages: 1026-1029)
- Research laboratories are defined as spaces for experimental investigation, testing, analysis, and controlled research activities
- Separation of research functions from public movement is emphasized
- Support spaces (instrument rooms, storage, service areas) are integral to research operation

2. Academic Facilities (TSS - Educational → College and University Facilities - Pages: 233-235)
- Academic buildings support teaching, learning, evaluation, and academic support
- Includes classrooms, lecture rooms, faculty offices, libraries, and student facilities

3. Exhibition Facilities (TSS - Cultural → Museums - Pages: 329-331)
- Museums and exhibition buildings serve public display, education, interpretation, and temporary exhibitions
- Distinction between public display zones and back-of-house zones

4. Tourism / Public Facilities (TSS - Governmental & Public Buildings - Pages: 702-704)
- Visitor-oriented buildings include reception, orientation, interpretation, food services, and amenities
- Designed for short-duration public occupancy

NBC & FIRE NORMS

1. EXIT REQUIREMENTS (EGRESS SYSTEM)

Number of Exits pp. 180-185

Minimum 2 exits required per floor for: Educational, Institutional, Assembly buildings

Exit Door Width pp. 200-205

Minimum clear width of exit door: 1000 mm
Exit doors must open in direction of travel
Sliding / revolving doors not permitted as exits

2. CORRIDORS & HORIZONTAL CIRCULATION (FIRE ACCESS)

Minimum Corridor Width pp. 180-190
- Institutional / Educational: 1.8 m
- Assembly buildings: 2.0 m recommended for crowd movement
- Maximum dead-end length: 6.0 m

3. TRAVEL DISTANCE

Condition Max Travel Distance pp. 185-195

Without sprinklers 30 m
With sprinklers 45 m

4. STAIRCASE NORMS (VERTICAL FIRE EGRESS)

Minimum Stair Width pp. 190-210
- Institutional / Educational / Assembly: 1.5 m
- Max riser: 150 mm
- Min tread: 300 mm
- Minimum Headroom: 2.2 m

5. FIRE TENDER ACCESS (SITE PLANNING LEVEL) NBC Part 3 & Part 4, pp. 150-170
- Minimum 6.0 m wide driveway
- Turning radius for fire tender required
- Access to all major building blocks

SPATIAL REQUIREMENTS & STANDARDS

Research Laboratories (TSS - Pages: 1029-1034)
- Typical laboratory module depth: **7.5 - 9.0 m**
- Laboratory bench width: **0.75 m**
- Minimum clear aisle between benches: **1.5 m**
- Instrument rooms separated from wet labs
- Service corridors provided alongside lab modules

Classrooms & Lecture Rooms (TSS - Educational → Classrooms: Pages: 190-195)
- Area per student: **1.2 - 1.5 sq.m**
- Typical classroom capacity: **30 - 60 students**
- Minimum aisle width: **1.2 m**
- Emphasis on unobstructed sightlines

Libraries (Academic & Research) (TSS - Libraries, Academic and Research: Pages: 257-274)
- Reading area: **2.5-3.0 sq.m per reader**
- Stack aisle width: **1.35-1.5 m**
- Individual study carrel width: **0.9-1.2 m**

Exhibition Galleries (TSS - Museums & Galleries - Pages: 334-339)
- Recommended gallery height: **4.5 - 6.0 m**
- Column-free spaces preferred
- Flexible layouts for temporary exhibitions
- Controlled lighting conditions required

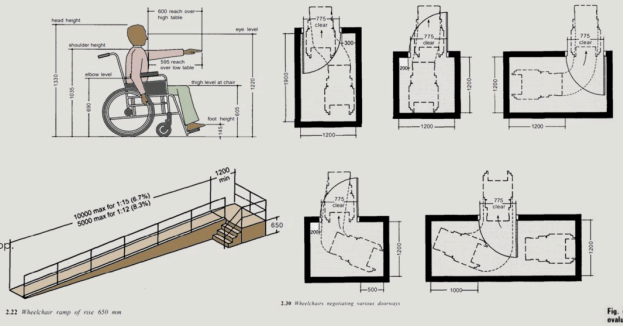
Guest Accommodation (Tourism) (TSS - Hotels & Motels - Pages: 870-899)
- Standard guest room size: **18 - 24 sq.m**
- Corridor width: **1.5 - 1.8 m**
- Separation of guest and service circulation

SERVICES & INFRASTRUCTURE

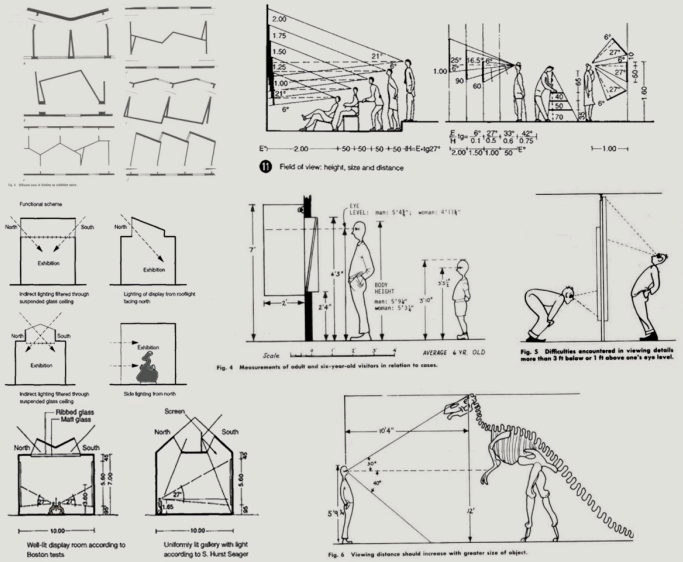
Ventilation & HVAC NBC 2016 Vol 2 – Part 8, Section 3 - Pages: 120–180
- Mechanical ventilation required for laboratories
- HVAC zoning recommended based on functional clusters
- Fresh air supply mandatory in occupied spaces

Electrical & Lighting NBC 2016 Vol 2 – Part 8, Section 2 - Pages: 85–120
- Separate electrical shafts recommended
- Emergency lighting required in corridors and exits

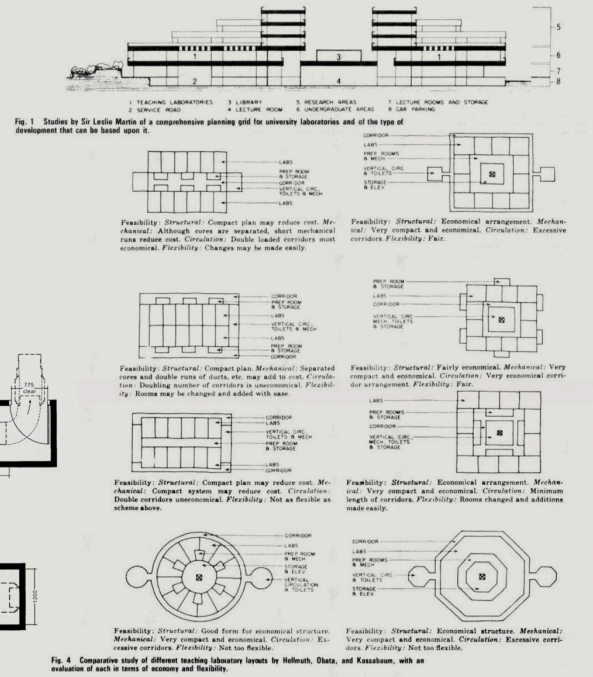
BARRIER FREE



EXHIBITION SPACE



RESEARCH LABS



RESEARCH QUESTIONS

- How can architecture reinterpret mining heritage to promote environmental awareness and sustainable reuse?
- How can a multidisciplinary campus integrate environmental research, education, and public engagement on post-mining land?
- How can architecture drive net-positive regeneration of reclaimed coal-mining land within a campus setting?

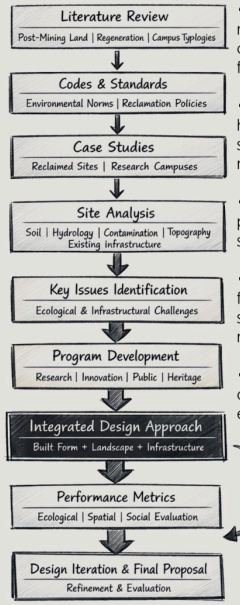
AIM

"To design a coal land reclamation, research & innovation campus addressing issues like coal regeneration processes, material innovation, challenges with post mine land along with performance metrics"

OBJECTIVES

- To respond to ecological and infrastructural challenges of reclaimed coal land through architecture
- To explore material innovation and adaptive construction for post-mining land.
- To integrate ecological regeneration processes into campus planning and landscape design.
- To establish performance metrics to evaluate environmental and spatial outcomes.

METHODOLOGY



INFERENCE

- Human anthropometry defines minimum spatial clearances and circulation widths across all campus functions.
- Research laboratories demand the highest spatial precision due to safety, equipment use, and multi-user movement.
- Academic spaces are governed primarily by seated dimensions, sightlines, and controlled circulation.
- Exhibition areas require wider, flexible circulation to support standing groups and public movement.
- NBC fire norms align circulation design with safe and efficient emergency egress.
- Public and service movements must be segregated to ensure safety and operational efficiency.
- Literature-based standards support scalable and future-ready campus planning.

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DESIGN BRIEF

INTRODUCTION

In response to escalating environmental degradation, this project proposes a **Regenerative Environmental Research and Learning Campus** that integrates scientific research, ecological restoration, and public engagement within a unified spatial framework.

The campus is envisioned as a living system - where built form, landscape, water, and soil function interdependently. Rather than imposing architecture on the site, the **design evolves from its topography, natural drainage patterns, and climatic conditions**, creating a responsive and climate-sensitive institutional environment.

By bridging research, education, and demonstration, the project aims to establish a model for future environmental campuses in India - one that not only studies ecosystems but actively restores and sustains them.

PROJECT VISION

To design an integrated campus that merges **advanced research infrastructure, academic learning environments, and a public-facing knowledge interface**, creating a spatial ecosystem where innovation, education, and community engagement coexist.
The project wishes to establish a controlled yet transparent institutional environment that:

- Facilitates **high-performance scientific research**
- Encourages **interdisciplinary academic exchange**
- Promotes **public awareness and experiential learning**
- Demonstrates **sustainable and regenerative design strategies**

SITE RESPONSE

The site strategy shall address:

- Climatic orientation and sun path
- Prevailing wind direction
- Noise buffering for research zones
- Public approach and institutional identity
- Segregated service access

Built form to respond through:

- Cluster formation
- Courtyard planning
- Landscape integration
- Controlled entry thresholds

PROJECT NEED

Chandrapur, characterized by coal-based industries, thermal power generation, and ecological sensitivity, faces severe environmental challenges including **air pollution, soil degradation, and water contamination.**

Despite its environmental vulnerability, there is a lack of integrated research and demonstration-based institutions addressing restoration and sustainable practices at a regional scale. This project responds by proposing a regenerative environmental research campus that directly engages with the climatic, industrial, and ecological realities of Chandrapur.

PROJECT FEASIBILITY

Chandrapur's industrial context, resource landscape, and growing environmental awareness make it a strategic location for such a campus.

The project supports regional research needs, promotes sustainable transition strategies, and can function as a demonstration model for coal-affected and ecologically stressed regions across central India.

DESIGN PRINCIPLES

Landscape-First Planning

The campus emerges from the site's natural topography and drainage patterns. Built forms are positioned to respect contours, preserve natural flows, and minimize land disturbance.

Zonal Hierarchy & Controlled Access

A clear gradient from public to restricted research areas ensures functional efficiency, security, and spatial clarity.

Climate-Responsive Architecture

Orientation, shading devices, courtyards, and thermal mass are used to reduce heat gain in Chandrapur's hot climate and enhance passive comfort.

Cluster-Based Organization

Programmatic blocks are arranged in clusters to promote interdisciplinary interaction while maintaining research privacy.

Built-Landscape Integration

Architecture and ecological systems function interdependently - outdoor testing areas, demonstration landscapes, and water systems become part of the spatial experience.

CHALLENGES

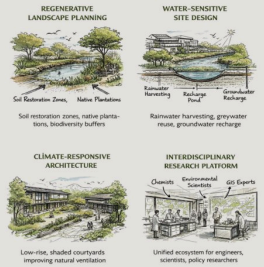
1. Industrial Pollution
Coal-based power plants and mining activities contribute to high particulate matter, degraded air quality, and environmental stress.

2. Soil Degradation
Fly ash disposal and extractive practices reduce soil fertility and ecological productivity.

3. Water Contamination & Poor Recharge
Surface runoff, industrial discharge, and inadequate recharge systems impact groundwater quality and availability.

4. Extreme Heat Conditions
Chandrapur experiences very high summer temperatures, intensifying heat island effects and reducing outdoor comfort.

5. Lack of Integrated Research Infrastructure
Existing systems lack a dedicated, interdisciplinary environmental research and demonstration campus for the region.



1. RESEARCH & INNOVATION ZONE

Space	Area (sqm)	Concurrent Users	Daily Footfall	Area/Person Logic
Wet Laboratories	700	60	70	11–12 sqm/person
Dry Laboratories	600	55	65	10–12 sqm/person
Specialized Research Labs	500	35	45	12–14 sqm/person
Pilot Plant / Experimental Labs	800	40	60	18–20 sqm/person
Clean Room	250	15	20	Controlled occupancy
Instrumentation Rooms	300	20	30	Equipment-dominant
Material Testing Lab	350	25	35	Vibration control
Prototyping & Fabrication Lab	500	35	45	Workshop format
Innovation / R&D Studios	400	30	40	Collaborative
Incubation Cells	250	20	30	Modular
Research Offices	350	40	45	8–9 sqm/person
Data & Simulation Centre	200	15	25	Server-supported
Documentation & Archive	150	10	15	Controlled
Research Meeting Rooms	180	25	40	Rotational use
Secure Storage	150	5	10	Restricted
Chemical Storage	120	5	8	Safety only
Gas Cylinder Yard	100	4	6	External
UPS / Electrical Room	150	5	8	Technical
Service Support Areas	250	10	15	Lab support
TOTAL AREA	6350			

2. ACADEMIC ZONE

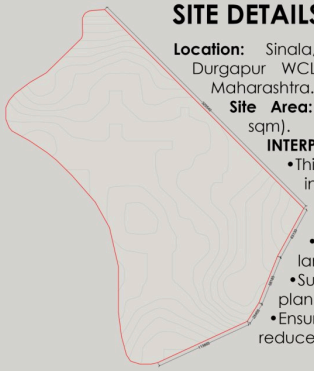
Space	Area (sqm)	Concurrent	Daily	Area Logic
Lecture Hall 1 (200 seats)	300	200	300	1.5 sqm/seat
Lecture Hall 2 (150 seats)	225	150	250	1.5 sqm/seat
Lecture Hall 3 (150 seats)	225	150	250	1.5 sqm/seat
Seminar Rooms (3 nos.)	360	120	200	3 sqm/person
Smart Classrooms (2 nos.)	300	100	160	2–2.5 sqm/person
Tutorial Rooms (4 nos.)	320	80	150	Small groups
Library & Knowledge Centre	1,000	250	350	3–4 sqm/seat
Digital Learning Labs	300	80	120	System-based
Faculty Offices	400	45	50	Fixed occupancy
Academic Administration	250	20	40	Visitor interaction
Student Commons	500	150	250	Informal seating
Discussion Lounges	250	80	150	Spill-out
Multipurpose Hall	400	200	300	Event-based
TOTAL AREA	4830			

3. EXHIBITION ZONE

Space	Area (sqm)	Concurrent	Daily	Area Logic
Permanent Gallery	500	120	200	4 sqm/person
Temporary Gallery	400	100	150	Flexible
Interactive Displays	300	70	120	Slower movement
Curatorial Office	100	6	8	Staff
Artifact Storage	150	4	6	Restricted
Prep / BOH	120	5	8	Service
TOTAL AREA	1570			

4. TOURISM & PUBLIC INTERFACE

Space	Area (sqm)	Concurrent	Daily
Interpretation Centre	400	120	250
Viewing Galleries	200	70	150
Café	450	150	300
Souvenir / Bookshop	150	40	120
Ticketing & Info	100	20	100
Public Toilets	150	50	—
TOTAL AREA	1450		



SITE DETAILS

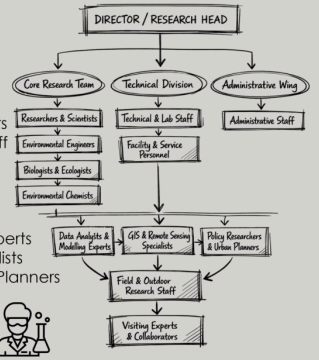
Location: Sinala, near Neela Pani, Durgapur WCL colony Chandrapur, Maharashtra.
Site Area: 17.07 Acres (~69,070 sqm).

INTERPRETATION

- This is a low density institutional development
- Suitable for a research campus
- Allows strong landscape dominance
- Supports regenerative planning model
- Ensures thermal comfort and reduced heat island effect

USER PROFILE

- Researchers & Scientists
- Technical & Lab Staff
- Research Scholars & Students
- Administrative Staff
- Visiting Experts & Collaborators
- Field & Outdoor Research Staff
- Facility & Service Personnel
- Environmental Engineers
- Environmental Chemists
- Biologists & Ecologists
- Data Analysts & Modelling Experts
- GIS & Remote Sensing Specialists
- Policy Researchers & Urban Planners



AREA PROGRAM

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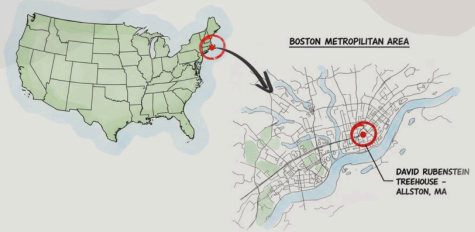
DAVID RUBENSTEIN TREEHOUSE DESKTOP CASE STUDY

INTRODUCTION



The David Rubenstein Treehouse serves as the primary gateway and conference hub for Harvard University's new **Enterprise Research Campus (ERC)**.
Designed by Studio Gang, the building is a masterclass in mass timber construction and sustainable urban integration, designed to foster collaboration between academia and the commercial sector.

Location: 100 Western Avenue, Allston, MA 02134, USA.

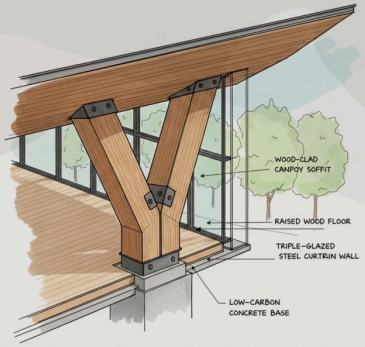
Site Context: It is situated within Phase A of the 14-acre Enterprise Research Campus in Allston, across the street from the Harvard Business School. It sits at a critical junction between the residential neighborhood of Allston and the evolving academic/innovation district.

CONCEPTUAL ANALYSIS

The project is conceived as a **vertical learning landscape**, where circulation becomes the primary social space. A continuous stair ramp system connects academic programs while **encouraging informal interaction, collaboration, and knowledge exchange through movement.**

- Circulation designed as a vertical social and learning spine
- **Continuous stair ramp system** connects all academic levels and encourages interaction
- Movement spaces **double as informal collaboration zones**
- Transparent, lightweight façade brings daylight and visually expresses the new intervention

FACADE TREATMENT



- **Minimal external intervention:** Existing building envelope largely retained to respect the historic structure.
- **Clear old–new contrast:** New insertion is visually distinct and expressed primarily on the interior.
- **Transparent internal façades:** Use of glass and open balustrades to maintain visual connectivity across levels.
- **Material hierarchy:** Warm timber elements contrast with the solid, heavier original façade.
- **Daylight-oriented treatment:** Openings and internal transparency allow daylight to penetrate deep into the building.

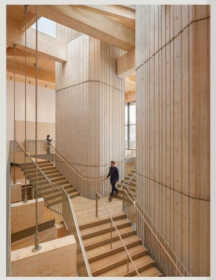
ZONING ANALYSIS

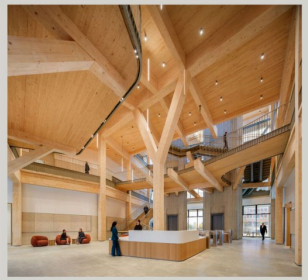
- **Public base (roots):** Ground level contains lobby, café, exhibition and public functions for open access
- **Circulation spine (trunk):** Central stair–ramp acts as the main social and movement core
- **Program clusters:** Meeting rooms, offices and academic spaces arranged around the spine
- **Canopy:** Mass-timber upper levels house event and collective learning spaces
- **Design strength:** Circulation doubles as a social space, encouraging interaction and collaboration

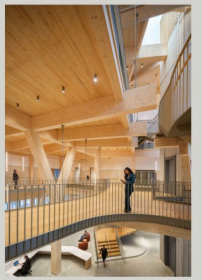
FIRST FLOOR PLAN

SECOND FLOOR PLAN

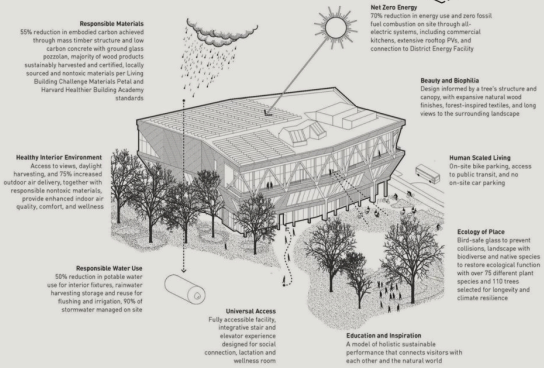
INTERIOR IMAGES







KEY ARCHITECTURAL FEATURES



INFERENCES

- **Circulation as architecture:** Design movement spaces as active social spines that support interaction, learning, and collaboration rather than only movement.
- **Clear vertical zoning:** Place public and shared functions at lower levels, with more focused academic or research spaces above for functional clarity.
- **Program clustering:** Organize offices, labs, and meeting spaces around circulation to promote visibility and informal exchange while maintaining separation.
- **Integrated informal spaces:** Embed breakout areas, seating, and discussion zones within circulation to extend learning beyond formal rooms.
- **Sustainable structure:** Use lightweight, low-carbon materials strategically to reduce environmental impact and improve spatial quality.
- **Collaboration-driven design:** Shape spaces to encourage interdisciplinary encounters through visual connections and shared platforms.

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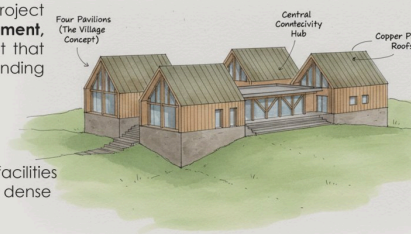
FRAUNHOFER RESEARCH CAMPUS DESKTOP CASE STUDY

INTRODUCTION

Fraunhofer Research Campus Waischenfeld, designed by Barkow Leibinger, is a precision driven research facility embedded within a **rural Bavarian landscape**. The project **emphasizes clarity, modularity, and low-impact development**, with architecture conceived as a functional instrument that supports advanced research while respecting the surrounding terrain.

Location: Waischenfeld, Bavaria, Germany
Client: Fraunhofer-Gesellschaft
Architects: Barkow Leibinger
Project Type: Research campus/experimental research facilities
Context: Built on previously altered / cleared land, not a dense urban site

CONCEPTUAL ANALYSIS



Conceived as a village-like research campus, with independent "house" volumes instead of a single monolithic block.

Programs are functionally separated yet spatially connected through shared outdoor and circulation spaces.

Architecture is embedded into the natural topography, minimizing visual and environmental impact.

Built form prioritizes precision, clarity, and efficiency, reflecting the nature of applied research.

Landscape acts as the primary connector, with buildings functioning as instruments within it.

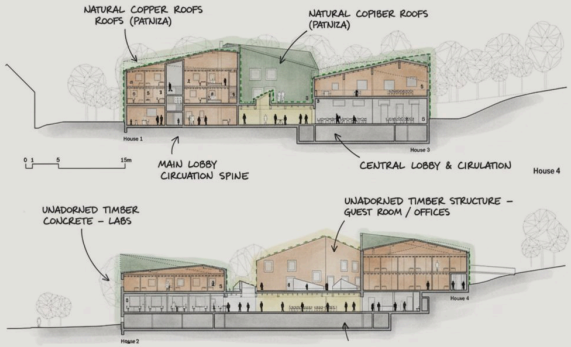
FLOOR PLANS

GROUND FLOOR PLAN

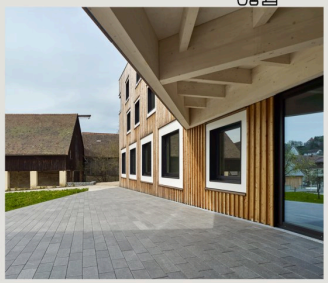
FIRST FLOOR PLAN

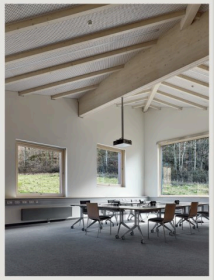
ZONING ANALYSIS

- The campus is **organized as a cluster of independent building volumes**, each assigned a specific function rather than a single consolidated block.
- Research and experimental spaces are placed in dedicated buildings **to maintain controlled environments and minimize disturbance.**
- **Support and technical functions are primarily located at lower levels**, acting as buffers and enabling efficient servicing.
- Administrative and office spaces are **positioned above or adjacent to support zones for close coordination** with research activities.
- **Guest and conference areas are located in separate**, quieter blocks to ensure privacy and comfort for visitors.
- Buildings are **connected through open circulation and landscape spaces**, making the landscape a key organizing element of zoning.



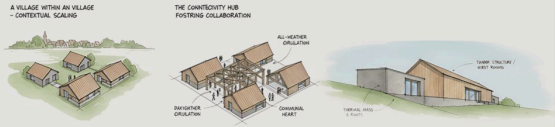
INTERIOR IMAGES





SITE IMAGES

KEY ARCHITECTURAL FEATURES



- Village-like cluster of buildings instead of a single consolidated block
- Independent "house" volumes allowing functional separation and flexibility
- Central shared spaces acting as social and circulation connectors
- Low-rise massing integrated with existing topography
- Clear zoning of research, support, administrative, and guest functions
- Simple, legible circulation with minimal internal corridors
- Material restraint emphasizing functionality, durability, and precision

INFERENCES

- Research functions perform better **when broken into smaller, independent volumes** rather than a single large block.
- **Functional separation with spatial proximity** allows technical research to remain controlled while enabling collaboration.
- **Support and technical spaces are best placed at lower levels** to buffer sensitive functions and simplify services.
- Administrative and guest functions should **occupy quieter, upper or peripheral zones to maintain privacy** and comfort.
- **Landscape can act as the primary connector**, reducing the need for enclosed circulation corridors.
- **Low-rise, terrain-following massing minimizes environmental impact on sensitive or reclaimed land.**
- Architecture as an instrument, not a landmark, **strengthens the functional and environmental logic of research campuses.**
- Clear zoning improves **operational efficiency and allows phased expansion** without disrupting ongoing research.
- **Decentralized planning improves adaptability**, allowing individual research blocks to **evolve or expand independently** without affecting the overall campus system.

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CSIR NEERI LIVE CASE STUDY

INTRODUCTION

CSIR–NEERI (National Environmental Engineering Research Institute) is a premier government-funded research campus focused on environmental science, engineering, and sustainability. Established in 1958, the institute functions under the Council of Scientific & Industrial Research (CSIR) and plays a critical role in advising national infrastructure, urban development, pollution control, and ecological restoration projects across India.

PROJECT DETAILS

- **Project Type:** Environmental research and institutional campus
- **Location:** Nagpur, Maharashtra
- **Campus Area:** ~125 acres
- **Built Form:** Low-rise, horizontally spread buildings
- **Primary Functions:** Research laboratories, pilot plants, administrative offices, academic facilities, residential quarters, and service infrastructure
- **Planning Approach:** Campus-in-park model with low built density and high landscape integration
- **Key Focus:** Sustainable water management, waste treatment, biodiversity conservation, and climate-responsive design

FUNCTIONAL ZONING

Core Research Zone
- Laboratories and pilot plants with controlled access
- Centrally located and buffered by green zones

Administrative Zone
- Director's office and administrative blocks
- Located near main entry for public access

Academic & Knowledge Zone
- Library, seminar and training facilities
- Positioned close to research areas

Residential Zone
- Staff housing and guest accommodation
- Separated from research areas for privacy

Utility & Services Zone
- Sewage treatment, water and service facilities
- Located at campus edge to minimise disturbance

Green & Buffer Zones
- Landscaped open spaces between functional zones

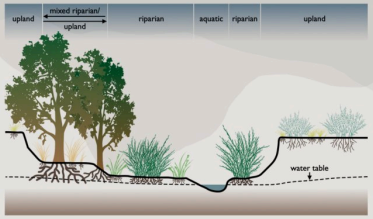
ECOLOGICAL STRATEGIES

- CSIR-NEERI functions as a living laboratory for environmental practices:
- On-site sewage treatment and water recycling
- Rainwater harvesting and groundwater recharge systems
- Extensive native plantation and biodiversity preservation
- In-house solid and hazardous waste research & treatment
- Heat mitigation through vegetation and low ground coverage

ARCHITECTURAL CHARACTER

- Predominantly low-rise buildings (G+1 to G+2)V
- Climate-responsive massing with shaded corridors and verandahs
- Functional, research-first architecture with minimal ornamentation
- Buildings positioned to allow airflow, daylight, and landscape continuity
- Architecture supports research efficiency rather than visual monumentality.

RESTORATION TECHNIQUES

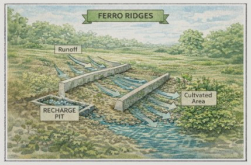


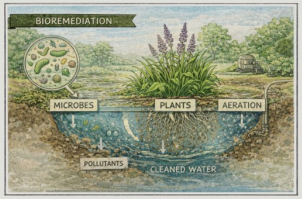
Riparian Management Plans (RMPs) are planning frameworks used to protect and manage land along rivers, lakes, and water channels. They define buffer zones, land-use controls, and ecological measures to maintain water quality, prevent erosion, and support biodiversity while allowing sustainable development.

What RMPs typically aim to do
- Protect water quality by filtering runoff, sediments, and pollutants
- Stabilise riverbanks to reduce erosion and flooding impacts
- Enhance biodiversity through native vegetation and habitat corridors
- Regulate human activity near water edges (setbacks, land use controls)

HRTS Pits (High Rate Transpiracy System)
- Engineered recharge pits or trenches designed to rapidly infiltrate stormwater into the ground.
- Used to reduce surface runoff, prevent waterlogging, and recharge groundwater, especially in campuses and large sites.
- Key role: Groundwater recharge + stormwater management.

Ferro Ridges
- Low ferrocement or RCC ridges/bunds constructed along contours or site edges.
- Slow down runoff, control soil erosion, and direct water towards recharge pits or planted areas.
- Key role: Runoff control + soil stabilisation.





Bio-remediation
Bioremediation is an environmental restoration process that uses microorganisms, plants, or natural biological processes to remove or neutralise pollutants from soil, water, or air. It works by breaking down contaminants into less harmful substances and is commonly used for treating polluted land, wastewater, and industrial sites.

SITE IMAGES

HRTS CONTROL UNIT

ACTIVE PIEZOMETER STORAGE

CONTROLLED CLIMATE

LAND RESEARCH LABS

ENG. SERVICE BLOCK

SOLAR PARK

SUSTAINIBILITY HEAD BLOCK

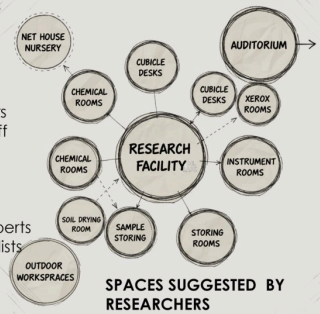
RESIDENTIAL BLOCKS

MESS ENTRY

ADMIN BLOCK

USER ANALYSIS

- Researchers & Scientists
- Technical & Lab Staff
- Research Scholars & Students
- Administrative Staff
- Visiting Experts & Collaborators
- Field & Outdoor Research Staff
- Facility & Service Personnel
- Environmental Engineers
- Environmental Chemists
- Biologists & Ecologists
- Data Analysts & Modelling Experts
- GIS & Remote Sensing Specialists
- Policy Researchers & Urban
- Planners



SPACES SUGGESTED BY RESEARCHERS

INFERENCES

- Site planning is **guided by existing ecology, topography, and drainage patterns.**
- Low-density, horizontal development supports **soil permeability and landscape continuity.**
- Functional zoning is clearly segregated while remaining connected through green buffers.
- Open spaces act as **active ecological systems** rather than residual areas.
- Water management systems are **integrated across the entire campus.**
- Regeneration processes are **continuous and monitored, not one-time interventions.**
- Utility and service zones are strategically placed to **minimise environmental disturbance.**
- The campus operates as a **live testing ground for regeneration strategies.**

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ENVIRONMENTAL REGENRATION SYSTEMS

PHYTOREMEDIATION

The Nature of the Challenge

•Dairy wastewater (DWW) is highly polluted due to lactose, whey, lactic acid, and fats (FOGs), resulting in very high Biochemical Oxygen Demand (BOD) that rapidly depletes oxygen in water bodies.

Biological Mechanism (Rhizosphere Action)

Rhizofiltration:
As wastewater enters the system, the dense fibrous root network physically traps suspended solids and particulate organic matter. This reduces turbidity and improves sedimentation efficiency.

Phytostimulation:
Plant roots release oxygen and organic compounds into the rhizosphere, creating localized aerobic zones that stimulate microbial activity even in otherwise low-oxygen wastewater.

Microbial Degradation:
Bacteria attached to the root surface metabolize lactose and whey through aerobic respiration, reducing organic load and lowering BOD and COD levels.

Phytoextraction:
Dissolved Nitrogen and Phosphorus are absorbed by plants and incorporated into structural biomass during rapid growth cycles.

Execution & System Design

•Constructed wetlands use gravel or sand media with horizontal or vertical flow to enhance root contact and oxygen transfer.

•Hydroponic systems grow floating plants directly in effluent for faster nutrient uptake.

•Hybrid systems use anaerobic digestion as pre-treatment before plant-based polishing.

Time & Economics

•Hydraulic Retention Time is typically 14-45 days, while floating plants like duckweed can double in 2-3 days under optimal conditions.

•Operational costs are lower than mechanical systems, and harvested biomass can be converted into biogas, compost, or animal feed, generating additional economic value.

MYCOREMEDIATION

The Nature of the Challenge

•Petroleum hydrocarbons, chlorinated solvents, and polycyclic aromatic hydrocarbons persist in soil due to their chemically stable ring structures. These compounds resist natural biodegradation and may remain toxic for decades.

Biological Mechanism (Enzymatic Decomposition)

Extracellular Enzyme Secretion:
White-rot fungi produce non-specific oxidative enzymes such as laccase, manganese peroxidase, and lignin peroxidase.

Oxidative Cleavage:
These enzymes break complex aromatic ring structures into smaller, less toxic molecules by destabilizing carbon-carbon bonds.

Mycelial Penetration:
Fungal hyphae physically penetrate soil micropores, expanding treatment zones beyond what bacteria alone can access.

Metal Immobilization:
Heavy metals are immobilized through adsorption onto fungal cell walls or converted into insoluble compounds, reducing mobility.

Execution & System Design

•In situ bioremediation treats contamination directly on site by introducing fungal inoculum into the soil, with proper aeration and moisture control to support enzyme activity.
•Bioaugmentation enhances the process by adding pre-adapted fungal strains to accelerate degradation in highly polluted soils.
•Solid State Fermentation uses agricultural residues as fungal substrates, converting contaminated material into stabilized compost-like products.

Time & Economics

•Petroleum hydrocarbon removal rates depend on fungal concentration and environmental conditions. Studies show significant reduction within approximately 60 days under optimized conditions.

•The process reduces excavation and transportation costs. However, site characterization and ongoing monitoring are required to ensure complete degradation without formation of toxic intermediates.

BIOCHAR PRODUCTION

The Nature of the Challenge

•Open burning of agricultural residues releases carbon dioxide, particulate matter, and volatile organic compounds. Simultaneously, soils in intensive agricultural regions experience declining organic carbon and poor moisture retention.

Thermochemical Mechanism (Pyrolysis)

Oxygen-Limited Heating:
Biomass is heated at moderate to high temperatures under limited oxygen conditions, preventing combustion and the formation of ash. Instead of burning, the material undergoes thermal decomposition, producing a stable carbon-rich biochar structure.

Thermal Decomposition:
Volatile compounds are released during heating, leaving behind a stable, carbon-rich matrix with a highly porous structure that enhances soil nutrient and water retention.

Porous Structure Formation:
The resulting biochar contains micro- and macro-pores that enhance cation exchange capacity and water retention.

Carbon Sequestration:
Carbon becomes chemically stable and can remain stored in soil for hundreds of years.

Execution & System Design

•Feedstock must be dried and size-reduced before entering pyrolysis units. Temperature and residence time control determine biochar quality and yield.

•Post-production, biochar must be cooled and handled carefully to prevent dust hazards. It is then incorporated into soil either by mixing or surface amendment.

Time & Economics

•Physical soil improvements occur immediately after application. Microbial colonization of biochar pores typically stabilizes within 3–6 months.

•Production costs vary significantly depending on scale, fuel inputs, and permitting requirements. Long-term economic viability improves when integrated with carbon credit systems.

CONSTRUCTED ECOSYSTEM

The Nature of the Challenge

•Urban development often replaces ecosystems that provided climate regulation, water filtration, carbon sequestration, and biodiversity support, while conventional buildings rarely compensate for these lost services.

Ecological Mechanism (Ecosystem Service Replication)

Baseline Ecological Mapping:
The pre-development ecosystem is identified and analyzed to quantify its service outputs, such as carbon sequestration, water filtration, climate regulation, and habitat support.

Performance Benchmarking:
Metrics such as annual carbon sequestration, stormwater absorption, and microclimate cooling are translated into architectural performance targets.

Integrated Biological Systems:
Green roofs, living walls, and constructed wetlands function as biological infrastructure embedded within built form.

Closed-Loop Resource Cycling:
Water, energy, and nutrient cycles are designed as localized, closed-loop systems, reducing dependence on centralized municipal infrastructure.

Execution & System Design

•Systems are integrated at building and site scale. Green roofs provide insulation and stormwater delay. Constructed wetlands treat greywater. Urban forestry moderates microclimates and enhances habitat connectivity.

Time & Economics

•Ecological systems generally mature within 2-5 years, after which they operate continuously and support environmental functions throughout the building's 50-100 year lifespan.

•Although initial capital costs are higher, long-term savings are achieved through reduced stormwater infrastructure demands, lower cooling loads, and improved occupant health and productivity.

PHYTOCAPPING

The Nature of the Challenge

•Conventional clay caps rely on impermeability but crack under desiccation and settlement. Cracking allows infiltration, generating toxic leachate and acid mine drainage.

Hydrological Mechanism (Evapotranspiration Control)

Soil Water Storage:
A thick, uncompacted soil layer functions as a water storage zone, retaining seasonal rainfall within the soil profile.

Root Zone Expansion:
Deep-rooted vegetation enhances soil porosity, improves structural stability, and reinforces slopes against erosion.

Evapotranspiration Pump:
Stored water is absorbed by plant roots and released into the atmosphere through evapotranspiration before it can percolate into the underlying waste layers.

Dynamic Adaptation:
Unlike rigid clay barriers, soil–vegetation systems respond dynamically to seasonal moisture changes without cracking or structural failure.

Execution & System Design

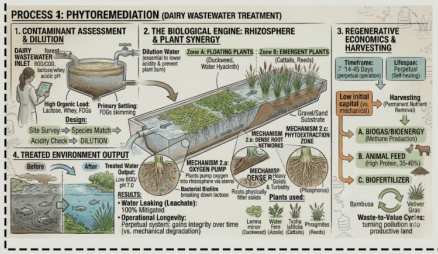
•Locally available soil is spread 1–2 meters thick over waste material. Pioneer grasses establish early erosion control, followed by deep-rooted tree species.

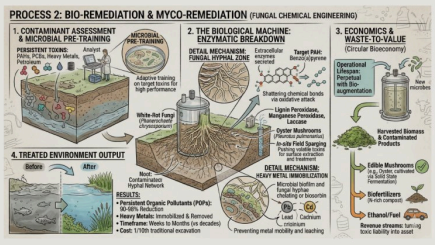
•Vegetation density and species selection are calculated based on peak rainfall and evapotranspiration rates.

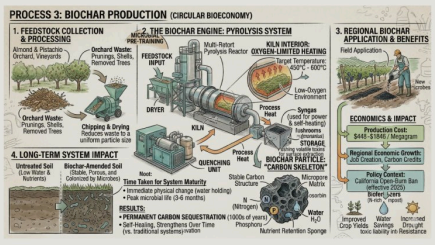
Time & Economics

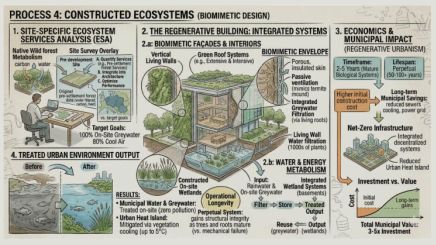
•Construction requires approximately 3–6 months. Full hydrological efficiency develops within 3–5 years as canopy cover increases.

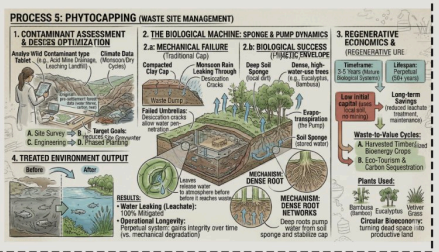
•Phytocaps are 40–60% cheaper than clay caps and require minimal long-term maintenance. Over time, the system strengthens biologically and may generate biomass value.











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COMPARITIVE ANALYSIS LITERATURE & CASE STUDIES

SR. NO.	DESIGN PARAMETER	LITERATURE STUDY	CSIR-NEERI	DAVID RUBENSTEIN TREEHOUSE	FRAUNHOFER RESEARCH CAMPUS	INFERENCE
1	SITE & CONTEXT RESPONSE	Research campuses should respond to climate, ecology, access and institutional role	Large, green campus responding to hot climate; inward-focused	Urban-edge, iconic insertion into campus expansion zone	Strategically located near industry & mobility networks	• Context is functional + institutional, not just climatic • National research campuses benefit from horizontal, landscape-led planning
2	CAMPUS SCALE & GRAIN	Scale should align with research type and user population	Low-rise, spread-out institutional fabric	Single compact vertical landmark	Multiple mid-scale modular buildings	• Large campuses need human-scaled clusters, not megablocks • Lab efficiency needs horizontality • Modular grain supports phased growth
3	PROGRAM NATURE	Labs, support spaces, admin & public areas must be clearly defined	Heavy lab + service dominance	Event, meeting, social focus	Research + pilot production + industry	• "Research" is not one typology • Program intent must be declared before planning • Requirements should be finalised before zoning
4	ZONING & ACCESS CONTROL	Gradation from public - restricted is critical	Strong separation & buffer zones	Highly open, minimal restrictions	Controlled but integrated zones	• Zoning communicates risk level and institutional openness • Labs demand buffer spaces (instrument, storage, service) • Public buildings can dissolve boundaries
5	SPATIAL ORGANIZATION	Clustering improves collaboration and efficiency	Functional lab clusters	Large open volumes & stacked spaces	Interlinked research & production clusters	• Clusters are better than linear blocks for research • Social buildings rely on visual connectivity & proximity
6	FLEXIBILITY & ADAPTABILITY	Research spaces must adapt to evolving technology	Limited by fixed lab services	High programmatic flexibility	High technical flexibility	• Flexibility is driven by service planning, not open plans • Lab buildings need phase-wise service corridors & spare capacity
7	STRUCTURE & MATERIALITY	Structure should support sustainability & adaptability	Conventional, service-friendly construction	Mass timber as identity & carbon strategy	Hybrid prefab systems	Structure is a design statement, not just engineering • Some specific materials suit people-centric spaces • Hybrid energy systems suit tech-heavy, evolving research
8	SUSTAINABILITY EXPRESSION	Sustainability should be integrated, not cosmetic	Demonstrated through research & landscape	Expressed through material choice	Demonstrated through tech systems	• Sustainability can be visible (form/material) and operational (systems) • Sustainability narratives can be clearly expressed through the integration of building services, thoughtful material selection, responsive landscaping strategies, and
9	USER EXPERIENCE	Balance focused work with interaction	Functional, work-driven experience	Strong experiential & social focus	Professional, industry-oriented	• Different users need different spatial quotient (intelectual, emotional, adverse, spiritual and creative) • Social spaces must be strategic nodes, not leftover lounges (path, edges, district, node and landmark)
10	PUBLIC INTERFACE	Research campuses should communicate knowledge	Outreach via demos & training	Building itself acts as interface	Industry-facing collaboration zones	• Architecture can act as a knowledge translator • GUI - Graphic User Interface • CLI - Command Line Interface • VUI - Visual User Interface • AR/VR - Augmented/Virtual Reality
11	INFRASTRUCTURE & SERVICES	Labs are service-dominated buildings	Very high service intensity	Minimal lab services	Industrial-grade infrastructure	• Smart and digital infrastructure • Safety and security systems • Inclusive and accesible design • Sustainable and eco-friendly campus design
12	GROWTH & FUTURE EXPANSION	Campuses should allow phased expansion	Organic growth over decades	Fixed iconic object	Highly scalable modular expansion	• Research campuses must allow future uncertainty - Net-Positive goals, Adaptive Re-use, AI data driven management, digital twins • Masterplanning matters more than individual buildings

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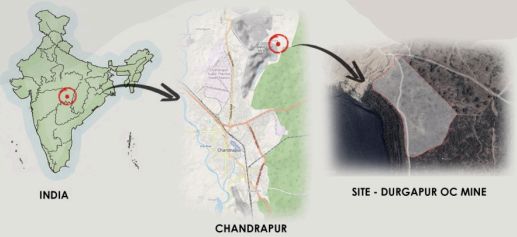
SITE ANALYSIS DURGAPUR OC MINES, CHANDRAPUR

LOCATION

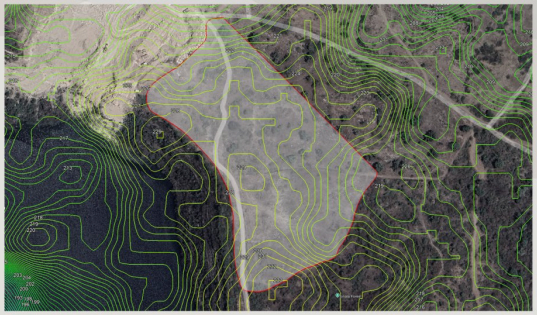


Geographic Location
- Latitude: 20.0184° N
- Longitude: 79.3314° E
- Region: Chandrapur district belt
- Site Area: 16 acres (~64,750 sq.m)

Regional Character
- Post-industrial landscape
- Extractive economy zone
- Ecologically stressed region

Surrounding Context
- West: Water body (Neela Paani)
- North-West: Open cast mining cut zone
- East & South: Scrub forest / semi-natural terrain
- Immediate Context: Low-density, non-urbanized landscape
- Nearest Settlement: Mining worker settlements (scattered)

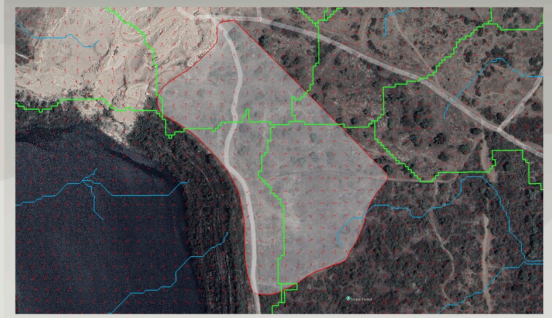
CONTOUR STUDY



The site sits at an elevation of approximately **230-250 m AMSL with an overall level variation of about 3-8 m.** It gently **slopes westward toward the Neela Paani** water body, while the north-western edge shows disturbed terrain due to adjacent mining activity.

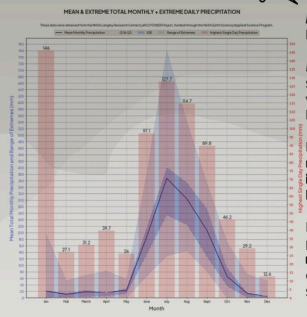
The **average slope ranges from 1:40 to 1:80**, making most of the land buildable, except for steeper patches near the mining edge. Natural drainage flows toward the western boundary, and with **1100-1200 mm annual rainfall, lower contours are more susceptible to monsoon runoff.**

MICRO-DRAINAGE ANALYSIS



The site exhibits a **natural westward drainage pattern**, with surface runoff directed toward the Neela Paani water body along the western boundary. **Multiple minor drainage channels originate from the eastern and north-eastern portions** of the site and converge toward the lower western edge.
The central portion of the site shows **relatively uniform flow direction, indicating a gentle gradient.** The western edge, being the lowest contour zone, acts as the primary discharge edge. Seasonal runoff during monsoon is expected to intensify along these visible natural channels.

CLIMATE ANALYSIS

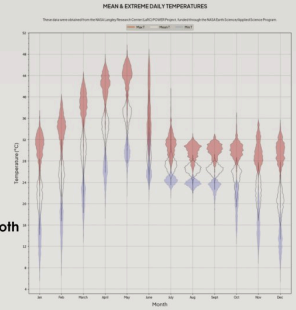


MEAN & EXTREME TOTAL MONTHLY + EXTREME DAILY PRECIPITATION

Mean monthly rainfall is highly seasonal, **peaking in July-August**, with very low **precipitation from November to May.**

Extreme single-day rainfall events reach **~127 mm in July and ~114 mm in August**, indicating short-duration, **high-intensity monsoon bursts.**

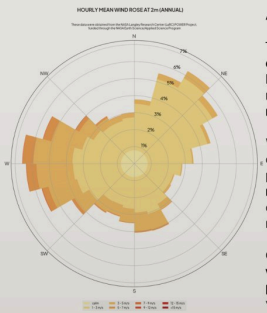
Rainfall variability is highest during peak monsoon months, **increasing runoff and drainage load**, while the extended dry season reflects strong seasonal water imbalance.



MEAN & EXTREME DAILY TEMPERATURES

Daily mean temperatures range from **approximately 20-37°C annually**, with peak summer maximums reaching **45-48°C** and winter minimums dropping to **10-14°C.**

Extreme daily maximums intensify during April-June, while seasonal trends indicate a **gradual rise in both daily minimum and maximum temperatures over time.**



ANNUAL WIND PATTERNS

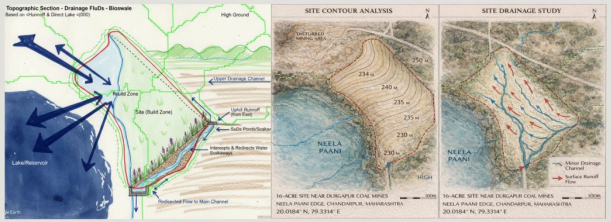
The annual wind rose indicates a **clear dominance of winds from the West and North-East quadrants**, with most hourly mean wind speeds ranging between **1-5 m/s.**

Western winds show higher frequency and consistency across the year, particularly **influencing summer and monsoon periods**. North-easterly winds are more prominent during winter months.

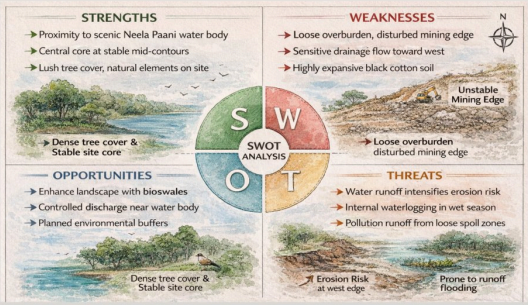
Overall, the **site experiences moderate wind intensities with a stable directional pattern**, supporting predictable passive ventilation strategies

CONNECTIVITY

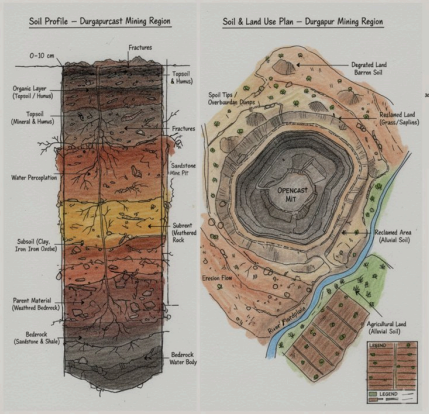
- Chandrapur Railway Station - 9.3kms
- Primary Health Center, Durgapur - 4.1kms
- Police Station - 8kms
- Indian Oil, Tukum - 5.7kms
- Zilla Stadium - 9.8kms
- Chandrapur Thermal Station - 7.2kms

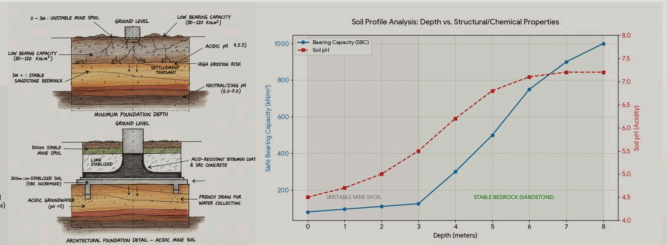


SWOT ANALYSIS



SOIL PROFILE ANALYSIS





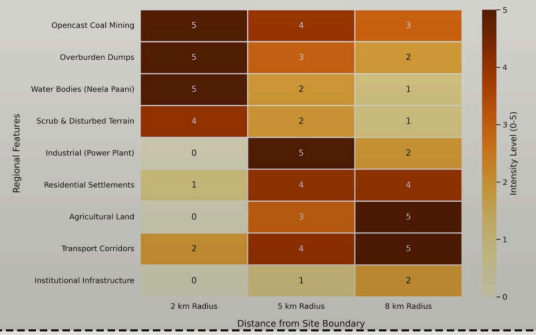
The site lies within the **Deccan Trap basalt region and is primarily covered with black cotton soil (expansive clay).** This soil exhibits significant **shrink–swell behavior due to seasonal moisture variation.** Areas near the mining edge may contain disturbed overburden material with inconsistent compaction.

The safe **bearing capacity generally ranges between 75–150 kN/m²**, subject to soil testing. The clay-rich composition results in low permeability and potential water retention during monsoon (1100–1200 mm annual rainfall).

At deeper levels, the **soil transitions into weathered basalt and coal-bearing sedimentary strata,** offering better stability when reached through appropriate foundation systems. **Careful load distribution and controlled foundation design** are essential for long-term performance.

NEIGHBORHOOD MATRIX

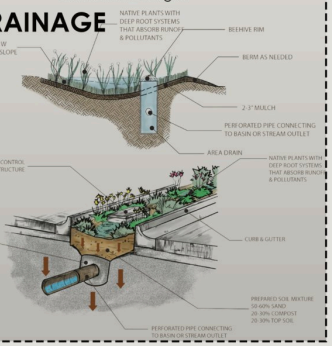
Regional Presence Matrix: Zonal Land-Use Intensity



SUSTAINABLE DRAINAGE

Sustainable drainage is a concept that includes **long term environmental and social factors** in decisions about drainage. It takes account of the **quantity and quality of runoff**, and the amenity and aesthetic value of surface water in the urban environment.

SUDs not just helps to manage and control the drainage issues but also provides **sustainable and eco friendly features like it helps increase green cover immensely.**



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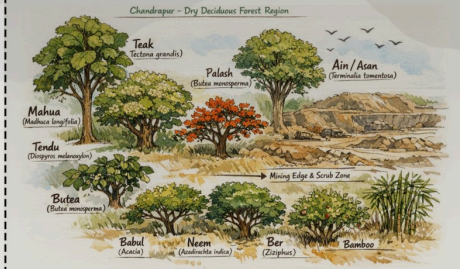
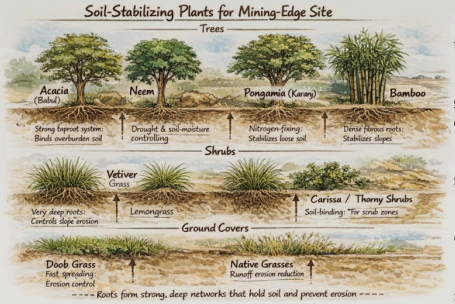
DURGAPUR OC MINES, CHANDRAPUR SITE ANALYSIS

FLORA ANALYSIS



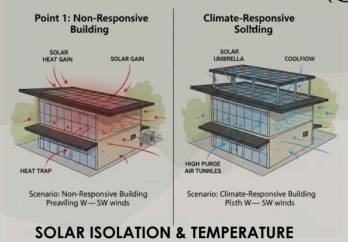


PRESENT SPECIES OF FLORA AROUND SITE

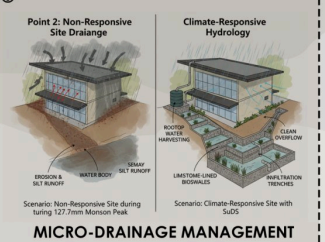
- **Teak (Tectona grandis) -** Dominant canopy species of the dry deciduous forest; drought-tolerant and seasonal.
- **Mahua (Madhuca longifolia) -** Hardy native tree adapted to dry climates with deep root systems.
- **Tendu (Diospyros melanoxylon) -** Common medium-sized forest species thriving in well-drained soils.
- **Ain / Asan (Terminalia tomentosa) -** Secondary canopy tree suited to dry, rocky terrain.
- **Palash (Butea monosperma) -** Drought-resistant tree known for seasonal flowering.
- **Neem (Azadirachta indica) -** Heat-tolerant species common in disturbed and semi-urban areas.
- **Babul (Acacia nilotica) -** Soil-binding tree suitable for degraded and mining-affected land.
- **Ber (Ziziphus spp.) -** Hardy scrub species adapted to arid conditions.
- **Bamboo (Dendrocalamus strictus) -** Useful for slope stabilization and wind buffering.

SITE IMAGES

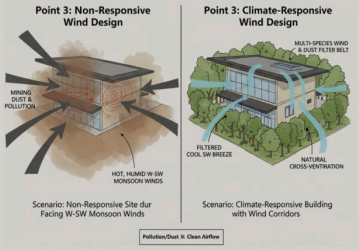
CLIMATIC INFERENCES



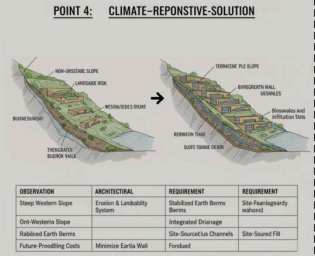
SOLAR ISOLATION & TEMPERATURE



MICRO-DRAINAGE MANAGEMENT



WIND FLOW OPTIMISATION



TOPOGRAPHY & CONTOUR PLAY

INFERENCES

- The site's natural **westward slope establishes a clear drainage hierarchy** that must guide master planning.**(STP, Rainwater collection-Services)**
- **Lower western contours are hydrologically sensitive and require controlled runoff and erosion management.(Low Height Buildings-Open field labs)**
- The central **moderately contoured plateau is most suitable for primary built** development.**(Main Buildings-Academic & admin)**
- Proximity to open-cast mining **demands ecological buffering and strategic mass placement** away from disturbed edges.**(Green Buffers)**
- Expansive black cotton soil necessitates **careful foundation design and avoidance of heavy point loads.(Helical Pile foundation)**
- High summer temperatures (45°C+) make **passive cooling strategies non-negotiable.(Building Orientation-NS)**
- Intense west solar exposure requires **façade protection and shaded transitional spaces.(Eg- Brise Soliel, Layered facade, brick jaali)**
- Strong seasonal solar availability offers **significant rooftop solar energy potential.**
- Highly seasonal monsoon rainfall with extreme daily events demands **integrated stormwater systems.**
- Natural catchment zones can be leveraged for **bioswales and groundwater recharge.(Stratergic Landscape)**
- Predominant West-Southwest winds **enable cross-ventilation but require dust filtration from mining side.(Green Buffer on West side)**

The site represents a post-extractive landscape, positioning the project as a regenerative ecological intervention.

SITE STUDY

SITE PARAMETERS

Site Area: **17.07 acres**
Proposed Road Width: **12m**

DRAINAGE STUDY

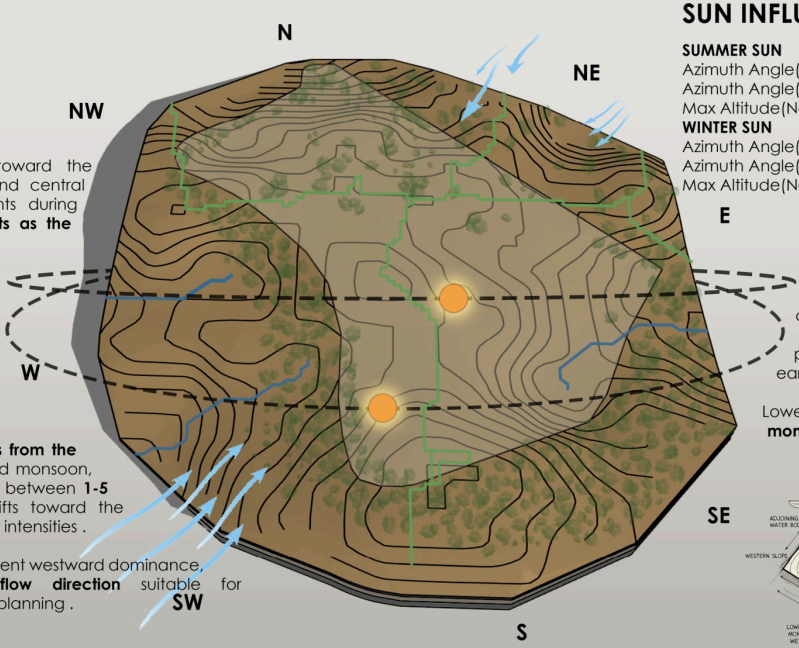
The site **drains naturally westward** toward the adjoining water body, with eastern and central areas functioning as runoff catchments during monsoon. The **lower western edge acts as the primary discharge zone.**

Integrating bioswales along natural drainage lines can **regulate runoff, enhance groundwater recharge, and reduce erosion**, while **vegetated swales help stabilize soil and filter surface water** before discharge.

WIND STUDY

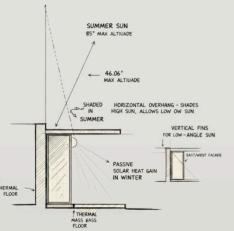
The site experiences **predominant winds from the West and South-West** during summer and monsoon, with average speeds generally ranging between **1-5 m/s.** During winter, wind direction shifts toward the North-East quadrant, with relatively lower intensities .

The annual wind pattern reflects a consistent westward dominance, establishing a **clear prevailing airflow direction** suitable for cross-ventilation and climate-responsive planning .



SUN INFLUENCE

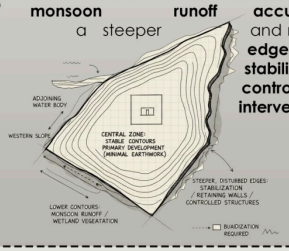
SUMMER SUN
Azimuth Angle(Sunrise): **73.08°**
Azimuth Angle(Sunset): **-69.02°**
Max Altitude(Noon): **85.06°**
WINTER SUN
Azimuth Angle(Sunrise): **114.55°**
Azimuth Angle(Sunset): **-114.99°**
Max Altitude(Noon): **46.28°**



CONTOUR STUDY

The contour study indicates a gentle **westward slope toward the adjoining water body**, defining the site's natural drainage direction. The **central zone exhibits moderate and stable contours,** making it suitable for primary built development with minimal earthwork.

Lower western contours are more **susceptible to monsoon runoff accumulation,** while a steeper and mining-disturbed **edges require stabilization and controlled structural intervention.**



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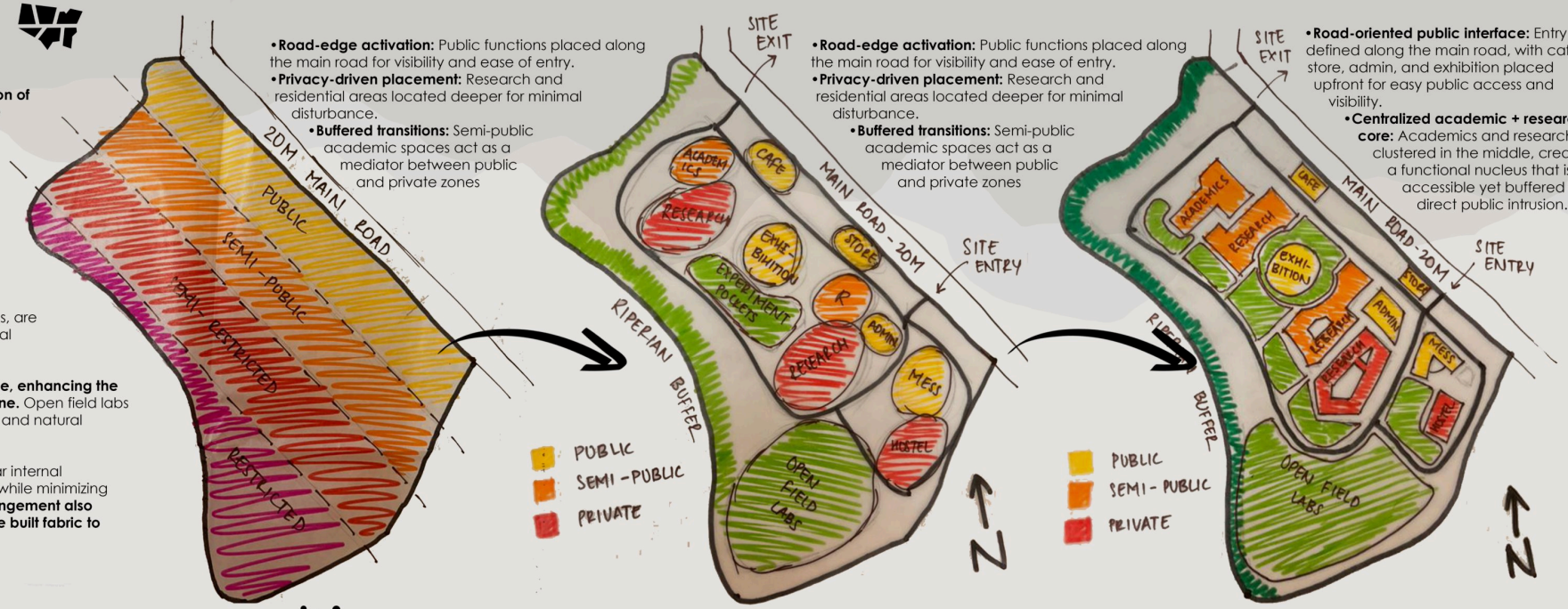
SITE ZONING

The site zoning is structured as a **clear gradation of accessibility and privacy**, shaped by both the site context and program requirements. The main road edge acts as the primary public interface, accommodating functions like exhibition, café, and admin for ease of access and visibility.

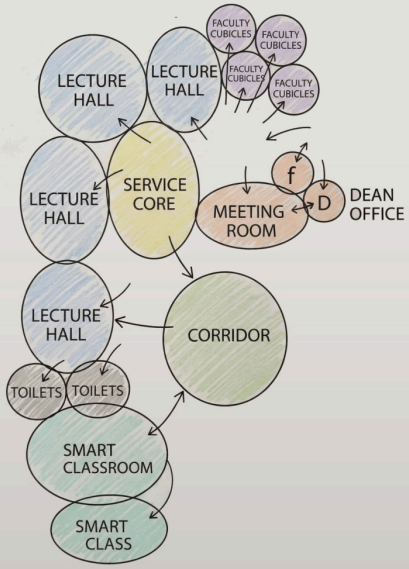
Moving inward, semi-public zones such as academic and collaborative research spaces **create a transitional layer that encourages interaction while maintaining controlled access.** The private zones, including research and residential areas, are placed deeper within the site to ensure minimal disturbance and a focused environment.

The **riparian buffer is retained as a natural edge, enhancing the microclimate and acting as an ecological spine.** Open field labs and landscape pockets further integrate built and natural systems.

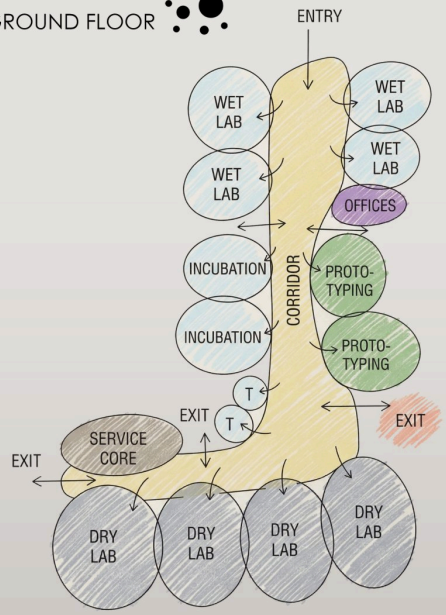
Additionally, the zoning is reinforced by a clear internal circulation loop, ensuring intuitive movement while minimizing conflict between user groups. The **spatial arrangement also responds to the site's natural form, allowing the built fabric to follow and respect its contours.**



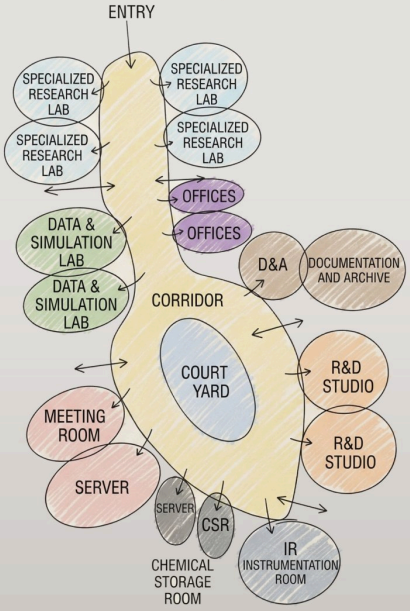
BUBBLE DIAGRAM GROUND FLOOR



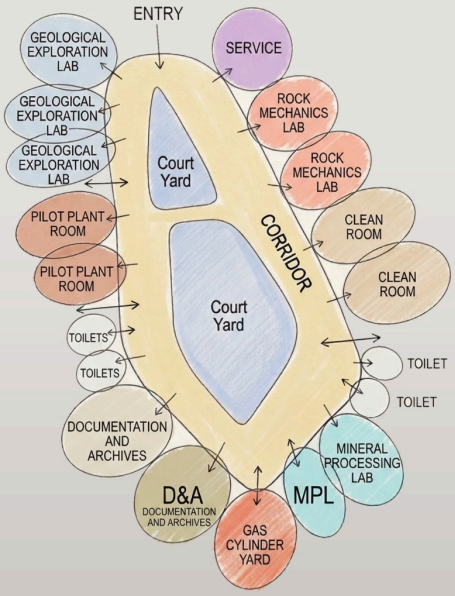
ACADEMIC BLOCK
BLOCK - A



RESEARCH BLOCK
BLOCK - B



RESEARCH BLOCK
BLOCK - C



RESEARCH BLOCK
BLOCK - D

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CONCEPT

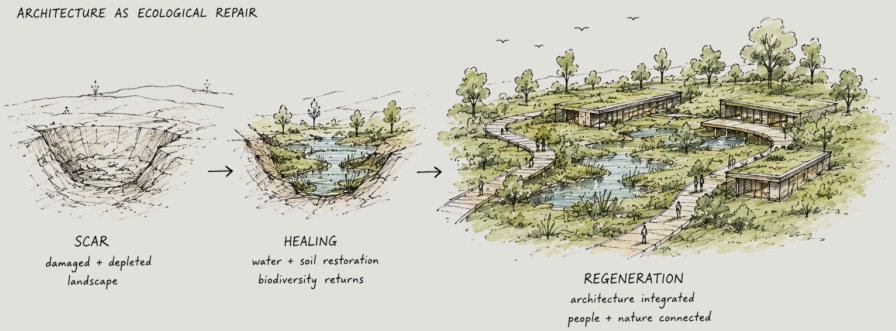
HEALING THE SCAR

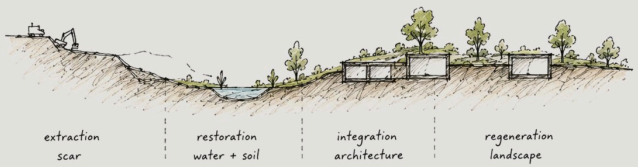
Architecture as Ecological Repair

The mined landscape is treated as a scar upon the earth, where architecture becomes a tool for **ecological healing rather than domination.** The campus is integrated within the terrain to **restore soil, water systems, biodiversity, and public connection with nature.**

Landscape becomes the primary planning framework, with wetlands, bioswales, plantation belts, and restoration zones shaping the site. **Buildings are lightly embedded into the land, following natural contours and water movement to minimize intervention.**

Fragmented low-rise forms, green roofs, earth berms, and elevated walkways create a seamless relationship between architecture and ecology. Over time, the built form appears reclaimed by nature, expressing regeneration and resilience.





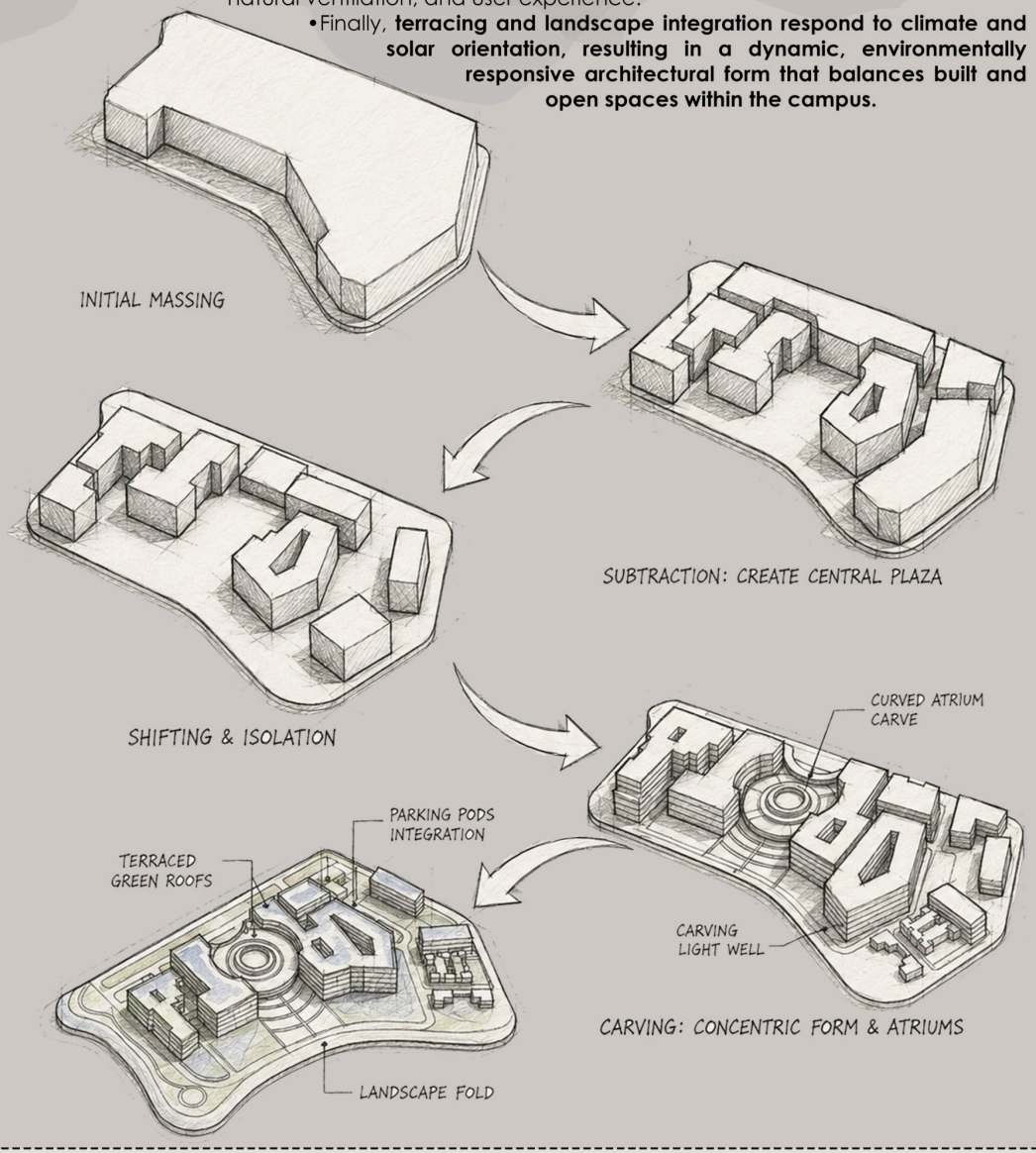
- Landscape-led
- Regenerative
- Embedded
- Porous
- Ecological Healing

MATERIAL LANGUAGE

DESIGN PRINCIPLES

FORM DEVELOPMENT

- The form development process evolves from a singular site-responsive mass into a cohesive architectural campus through a series of strategic interventions. **Initial subtraction creates a central public plaza that establishes spatial hierarchy and interaction zones.**
- The built mass is then fragmented and shifted to improve connectivity, daylight access, and functional zoning across the site.
- Further carving introduces atriums and concentric spatial elements that enhance visual continuity, natural ventilation, and user experience.
- Finally, **terracing and landscape integration respond to climate and solar orientation, resulting in a dynamic, environmentally responsive architectural form that balances built and open spaces within the campus.**



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SITE PLAN SCALE 1:750

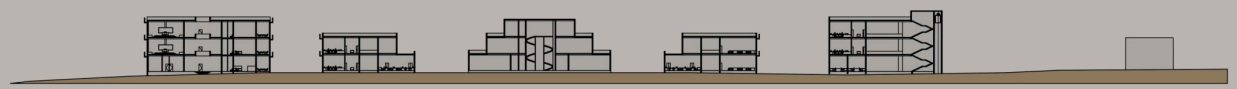
AREA STATEMENT

1. SITE AREA - **69117 SQ.M/ 17 ACRES**
2. GROUND COVERAGE - **14346 SQ.M**
3. BUILT UP AREA - **29115 SQ.M**
4. ACHIEVED GROUND COVERAGE - **20.75%**

LEGEND

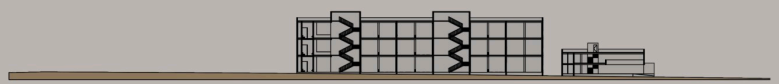
1. MAIN ENTRY/EXIT
2. SECONDARY ENTRY/EXIT
3. ON SITE PARKING
4. SOVENIER SHOP
5. CAFE
6. ADMIN BLOCK
7. ACADEMIC BLOCK - A
8. RESEARCH BLOCK - B
9. EXHIBITION BLOCK
10. RESEARCH BLOCK - C
11. RESEARCH BLOCK - D
12. MESS
13. STAFF/WORKERS RESIDENCY
14. STUDENT HOSTEL
15. GUEST HOUSES
16. VC RESIDENCE
17. PRESIDENT RESIDENCE
18. RIPERIAN BUFFER
19. OPEN FIELD EXPERIMENTATION LABS
20. CIRCULAR LANDSCAPE
21. RESTORATIVE GARDEN
22. OPEN COURTYARD
23. COLLABORATIVE COURTYARD
24. JOGGING/CYCLING TRACK

BUILDING SECTIONS



SITE SECTION AA'

SITE SECTION BB'



SITE SECTION CC'

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GROUND FLOOR SITE PLAN SCALE 1:750

AREA STATEMENT

1. SITE AREA - **69117 SQ.M/ 17 ACRES**
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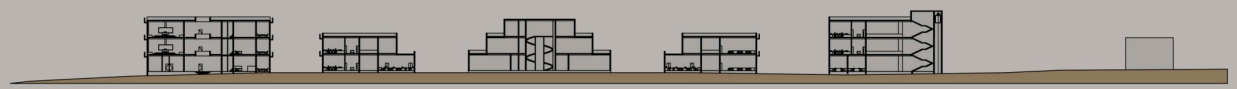
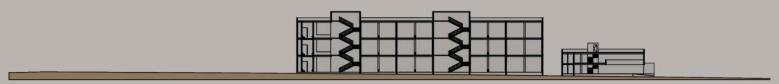
5. CAFE
6. ADMIN BLOCK
7. ACADEMIC BLOCK - A
8. RESEARCH BLOCK - B
9. EXHIBITION BLOCK
10. RESEARCH BLOCK - C
11. RESEARCH BLOCK - D
12. MESS
13. STAFF/WORKERS RESIDENCY
14. STUDENT HOSTEL
15. GUEST HOUSES
16. VC RESIDENCE
17. PRESIDENT RESIDENCE
18. RIPERIAN BUFFER
19. OPEN FIELD EXPERIMENTATION LABS
20. CIRCULAR LANDSCAPE
21. RESTORATIVE GARDEN
22. OPEN COURTYARD
23. COLLABORATIVE COURTYARD
24. JOGGING/CYCLING TRACK

BUILDING SECTIONS



SITE SECTION AA'

SITE SECTION BB'



SITE SECTION CC'

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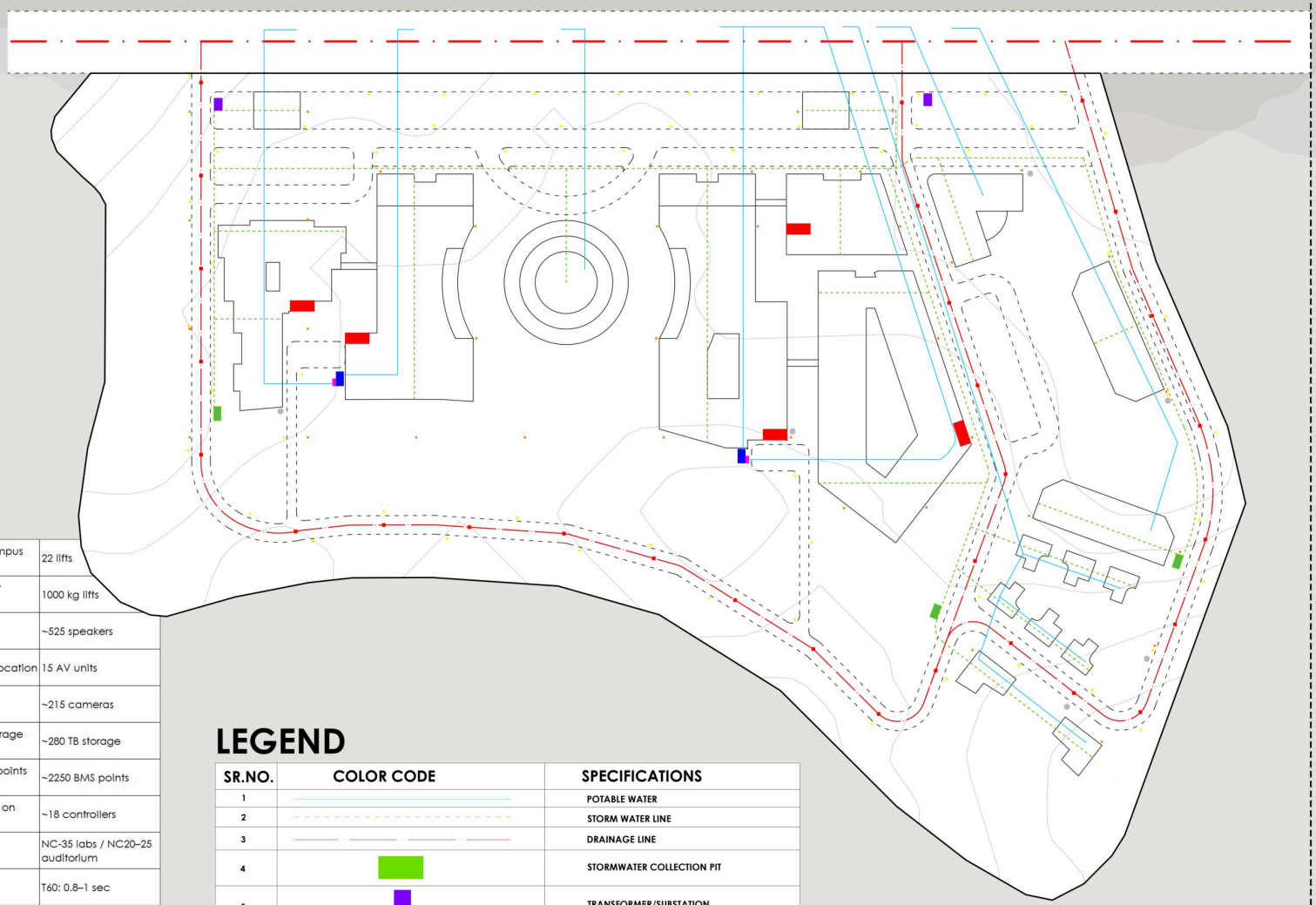
SERVICE PLAN SCALE 1:750



SERVICE CALCULATIONS

SYSTEM	KEY CALCULATION / INFERENCE	FINAL PROVISION
HVAC	Cooling load derived from occupancy + lighting + equipment + solar gains	~700 TR cooling load
	Climate-based operation reduced to ~4.7 hrs/day	4 × 200 TR chillers (1 standby)
	Chilled water flow based on ΔT method	~115–120 L/s
Water Supply	Domestic + HVAC demand combined	~97.5 KLD total demand
	STP greywater reuse offsets freshwater demand	~40–45 KLD freshwater demand
	~80% wastewater recovery assumed	~59 KLD greywater recovery
Fire Fighting	NBC ordinary hazard calculations	~28 L/s fire demand
	Flow × duration storage logic	~75 KL fire tank
	Coverage radius based layout	Hydrants at 30–45 m spacing
Solar PV	Rooftop area × panel density	~600 kWp system
	PSH-based annual generation	~990,000 kWh/year
	400 W panel assumption	~1500 panels
Drainage & STP	80% water demand converted to sewage	~78 KLD sewage
	+20% buffer for treatment sizing	~95 KLD STP
	Rational runoff method used	~200 L/s rooftop runoff
Electrical	HVAC + lighting + equipment + UPS loads combined	~2366 kW connected load
	Demand factor applied	~1775 kW demand load
	PF-based transformer sizing	2 × 1250 kVA transformers
DG & Backup	Essential load restricted to ~40% only	2 × 450 kVA DG sets
	Battery sized for critical systems	~300 kWh backup
	8-hour autonomy considered	~1500 L fuel storage

Lifts	Low-rise distributed campus logic	22 lifts
	Accessibility + stretcher compliance	1000 kg lifts
PA & AV	PA density based on circulation areas	~525 speakers
	Teaching-space AV allocation	15 AV units
CCTV & Security	Perimeter + circulation surveillance	~215 cameras
	30-day compressed storage estimate	~280 TB storage
IBMS / BMS	Integrated monitoring points across services	~2250 BMS points
	DDC distribution based on point capacity	~18 controllers
Acoustics	ASHRAE + ISO acoustic standards	NC-35 labs / NC20–25 auditorium
	Reverberation control calculations	T60: 0.8–1 sec
Rainwater Harvesting	Rainfall × runoff coefficient method	~39 ML/year potential
	Recharge-first hydrological strategy	Swales + recharge pits
Organic Waste & Biogas	Organic waste converted to biogas	~145 kg/day waste
	Retention-time-based digester sizing	~7 m³ digester
Resource Balance	Rainwater exceeds annual demand	Water positive system
	Solar offsets operational energy	~30–35% energy offset

LEGEND

SR.NO.	COLOR CODE	SPECIFICATIONS
1		POTABLE WATER
2		STORM WATER LINE
3		DRAINAGE LINE
4		STORMWATER COLLECTION PIT
5		TRANSFORMER/SUBSTATION
6		SUMP
7		FIRE FIGHTING WATER TANK
8		MUNCIPAL SEWER LINE
9		FIRE HYDRANT
10		STREET LIGHT
11		GARBAGE COLLECTION POINT
12		INSPECTION CHAMBER
13		FIRE ESCAPE CHAMBER

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VIEWS

BIRD'S-EYE VIEW

LIVE RESEARCH

SITE TOP VIEW

OFL AND RESEARCH BLOCKS

EXHIBITION WITH RESEARCH - B

EXHIBITION BLOCK

SITE VIEW AT ENTRY

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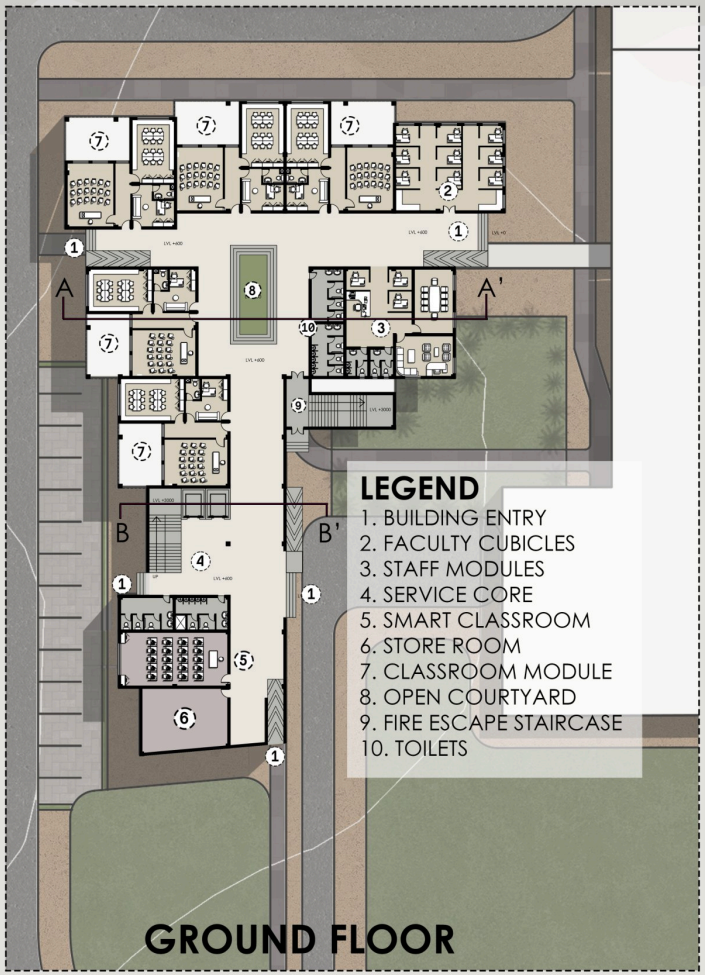
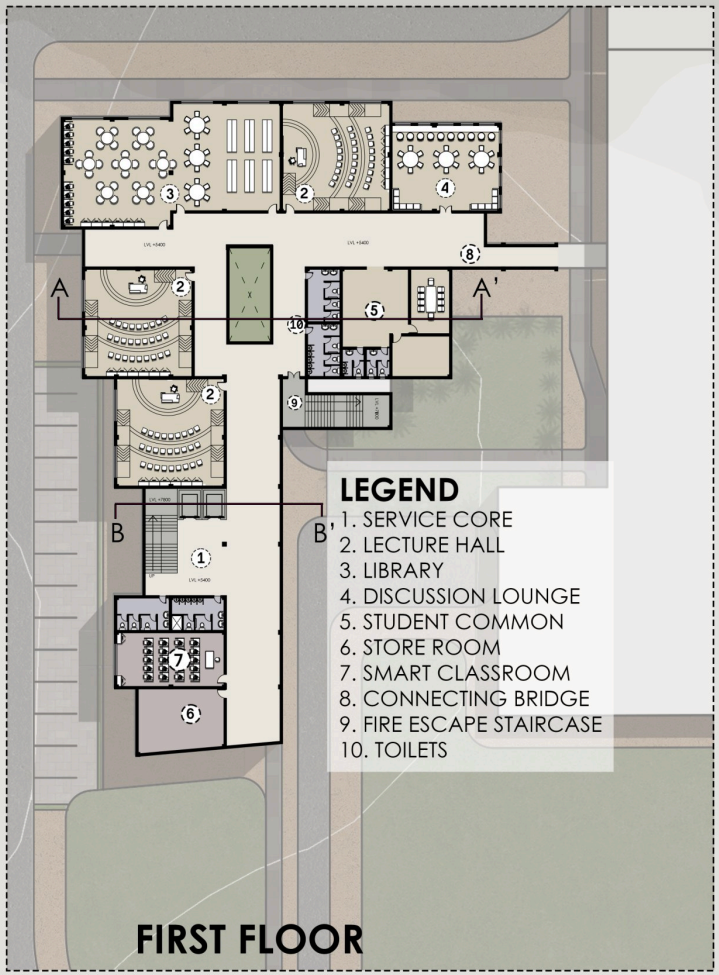
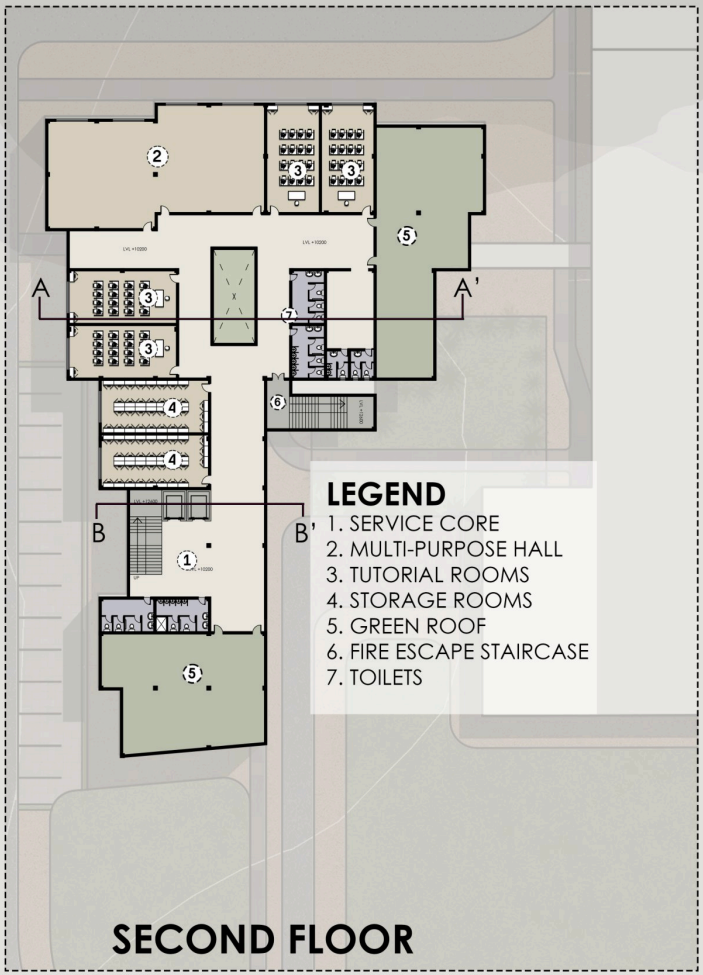
ACADEMIC BLOCK BLOCK-A (SCALE 1:250)



GROUND FLOOR

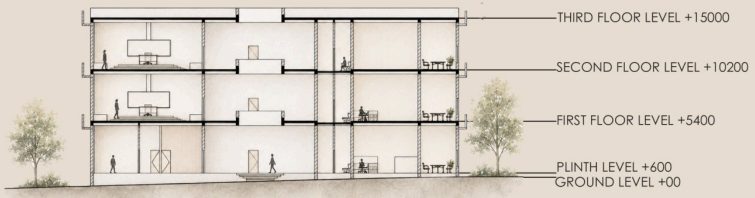


FIRST FLOOR

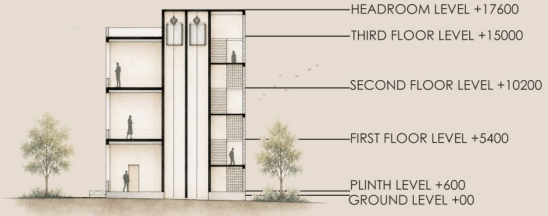


SECOND FLOOR

BUILDING SECTIONS



SECTION AA'



SECTION BB'

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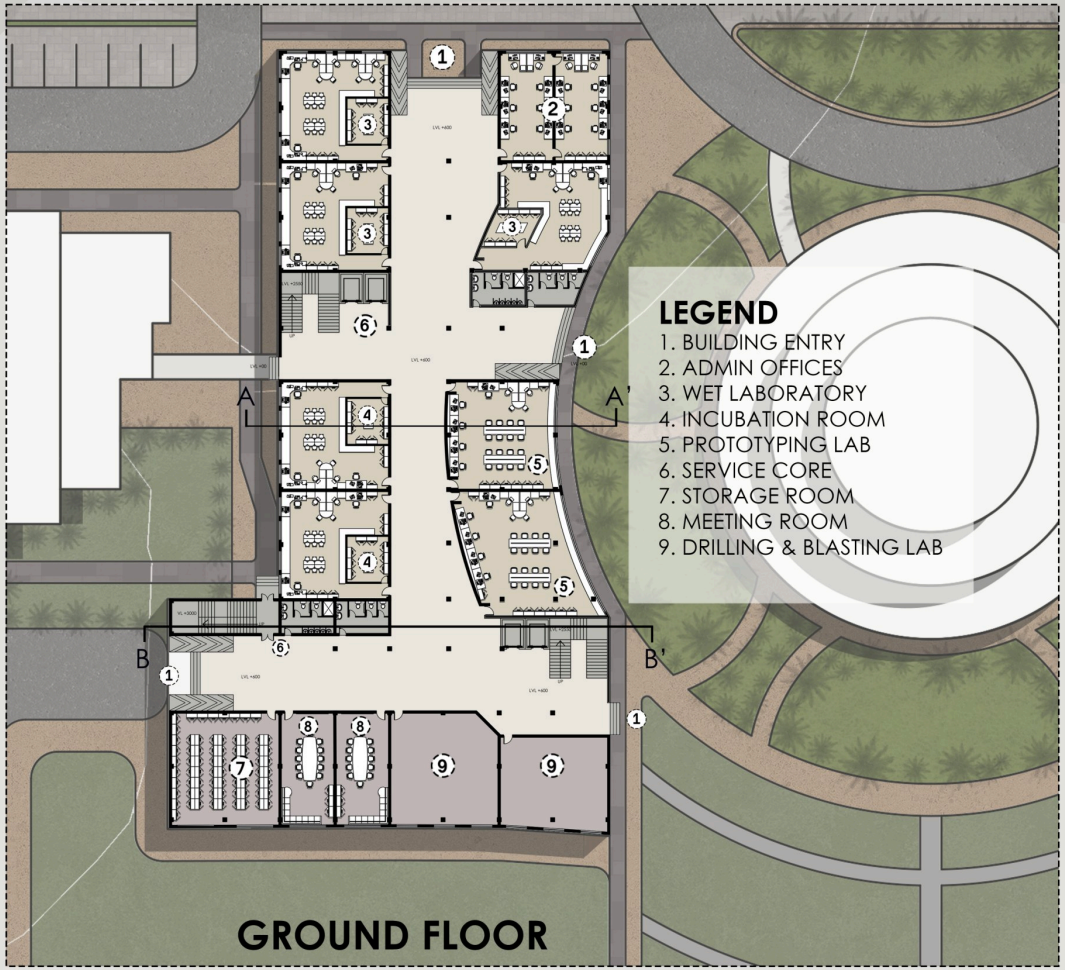
RESERACH BLOCK BLOCK-B (SCALE 1:250)



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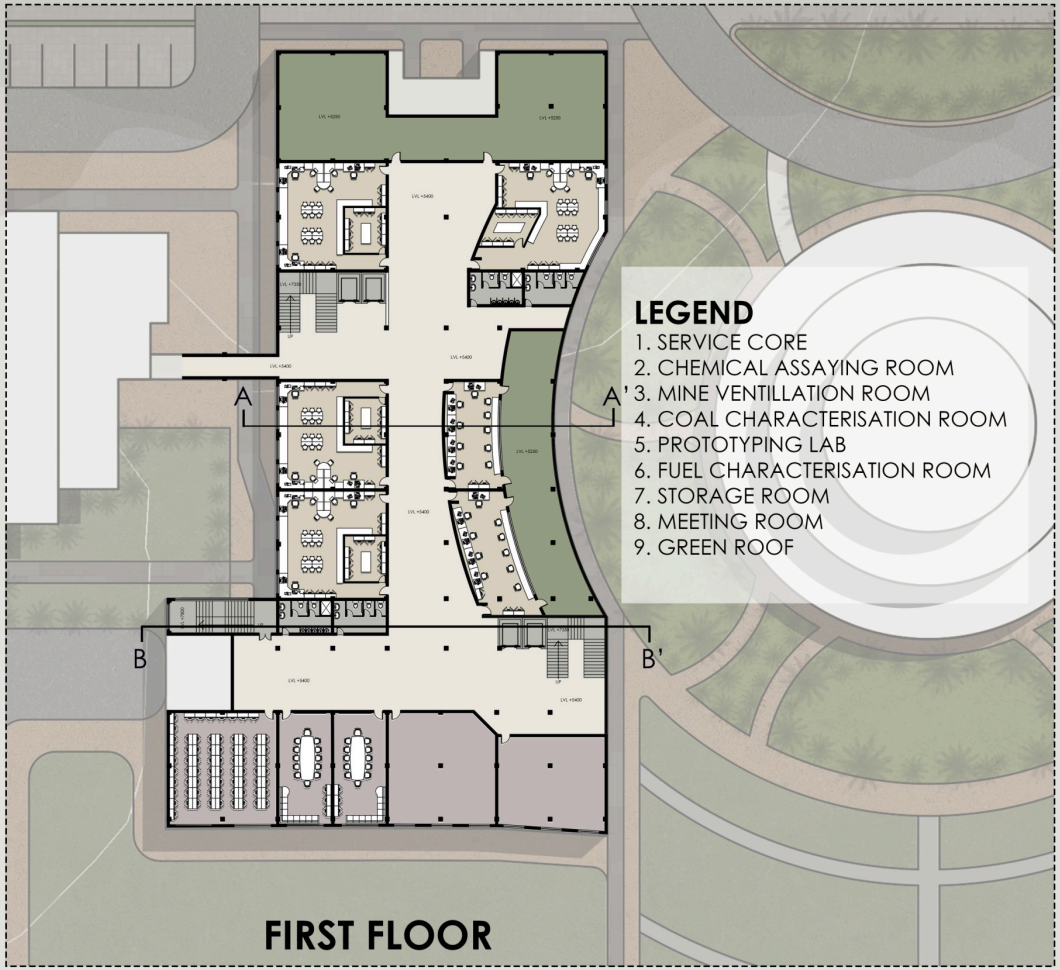
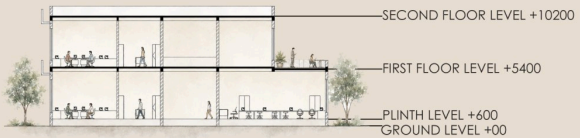
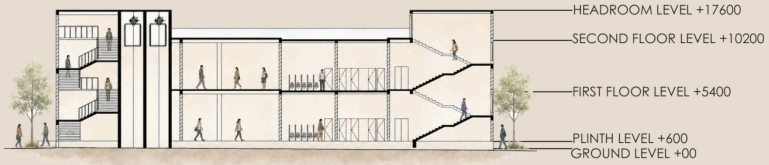


FIRST FLOOR

BUILDING SECTIONS



SECTION AA'



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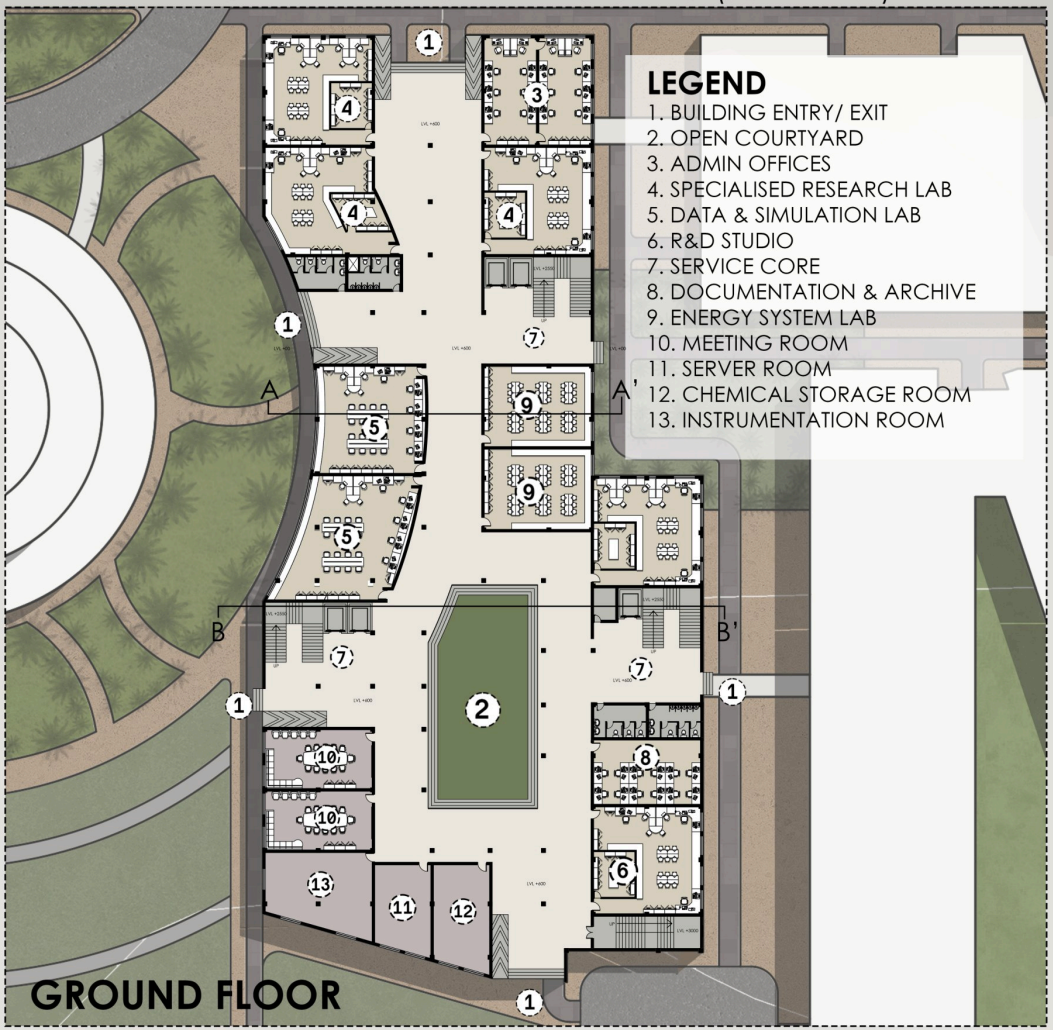
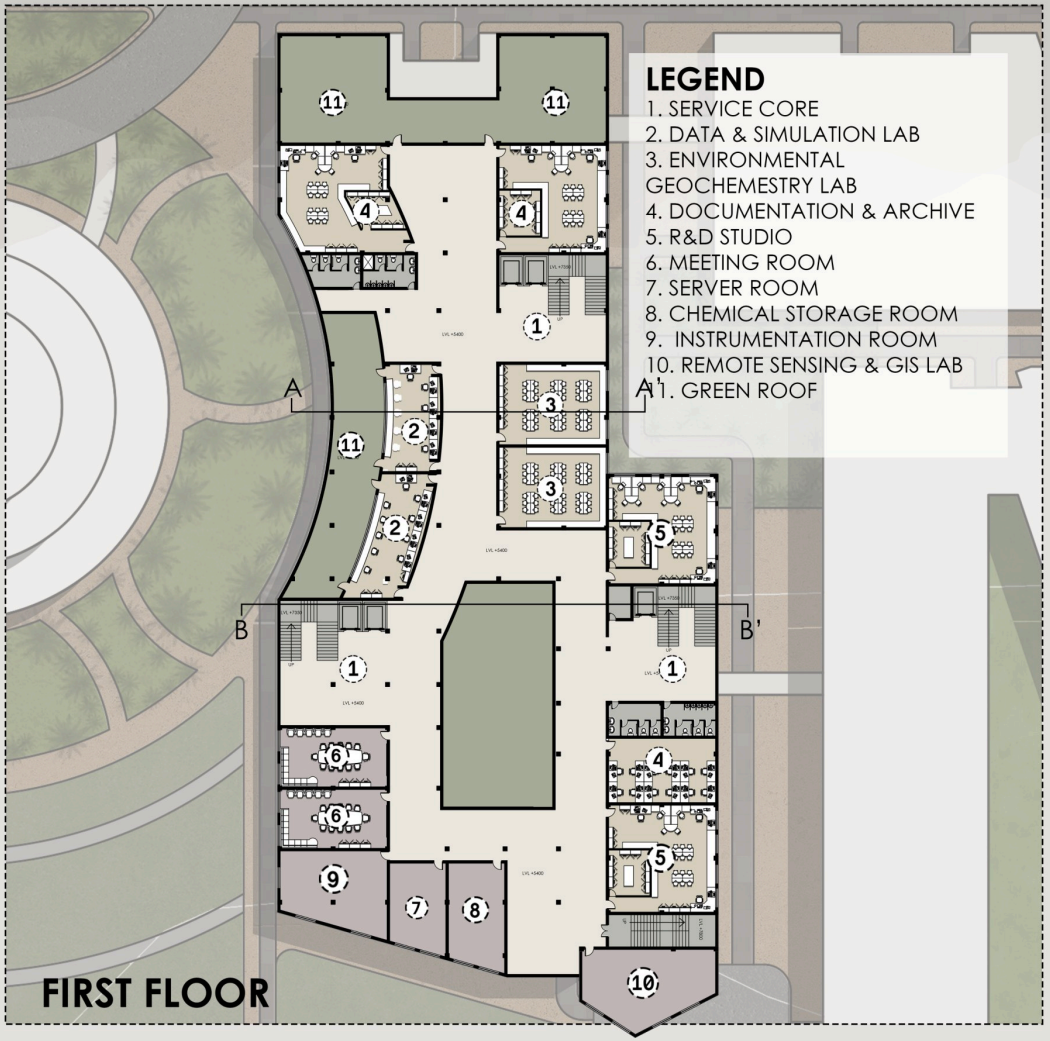
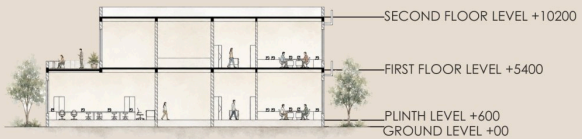
RESERACH BLOCK BLOCK-C (SCALE 1:250)





BUILDING SECTIONS



SECTION AA'

SECTION BB'

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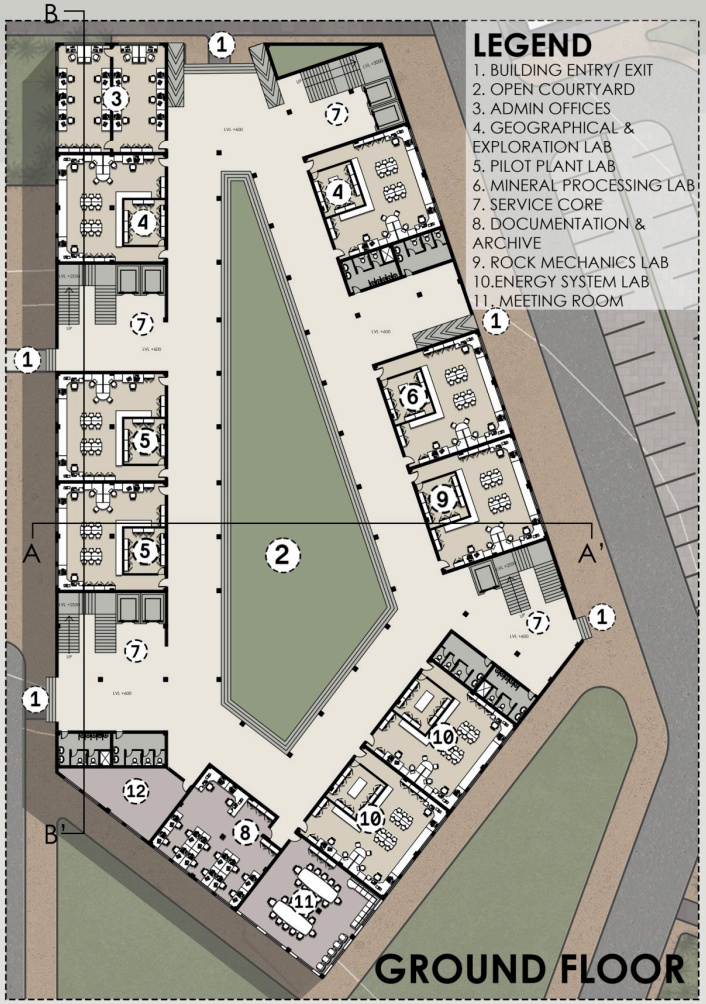
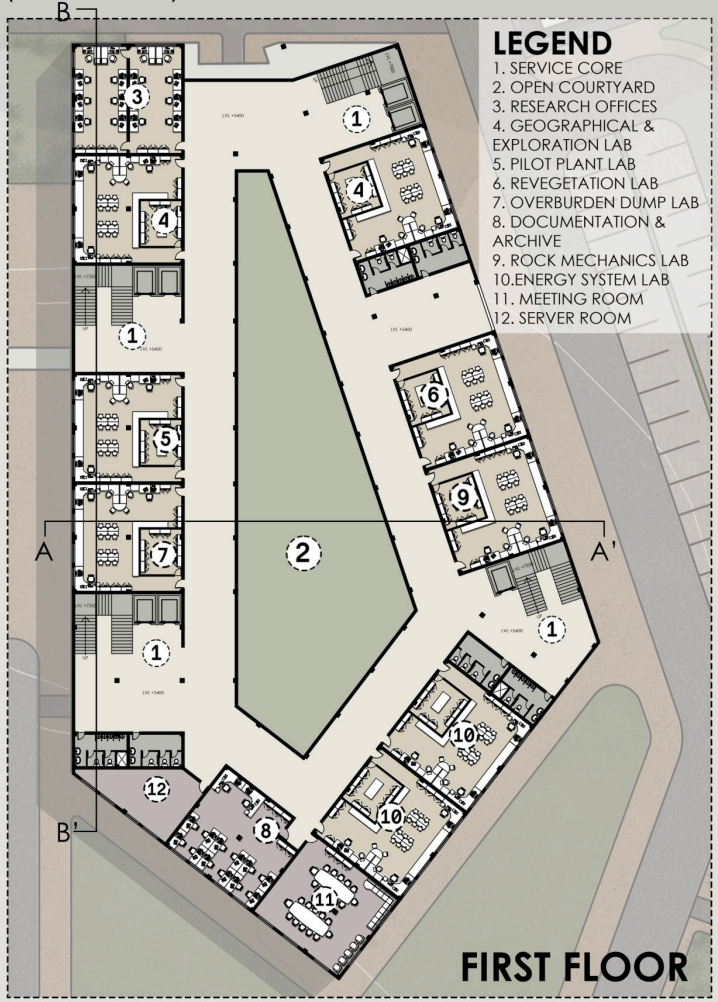
RESERACH BLOCK BLOCK-D (SCALE 1:250)



LEGEND

1. BUILDING ENTRY/ EXIT
2. OPEN COURTYARD
3. ADMIN OFFICES
4. GEOGRAPHICAL & EXPLORATION LAB
5. PILOT PLANT LAB
6. MINERAL PROCESSING LAB
7. SERVICE CORE
8. DOCUMENTATION & ARCHIVE
9. ROCK MECHANICS LAB
10. ENERGY SYSTEM LAB
11. MEETING ROOM

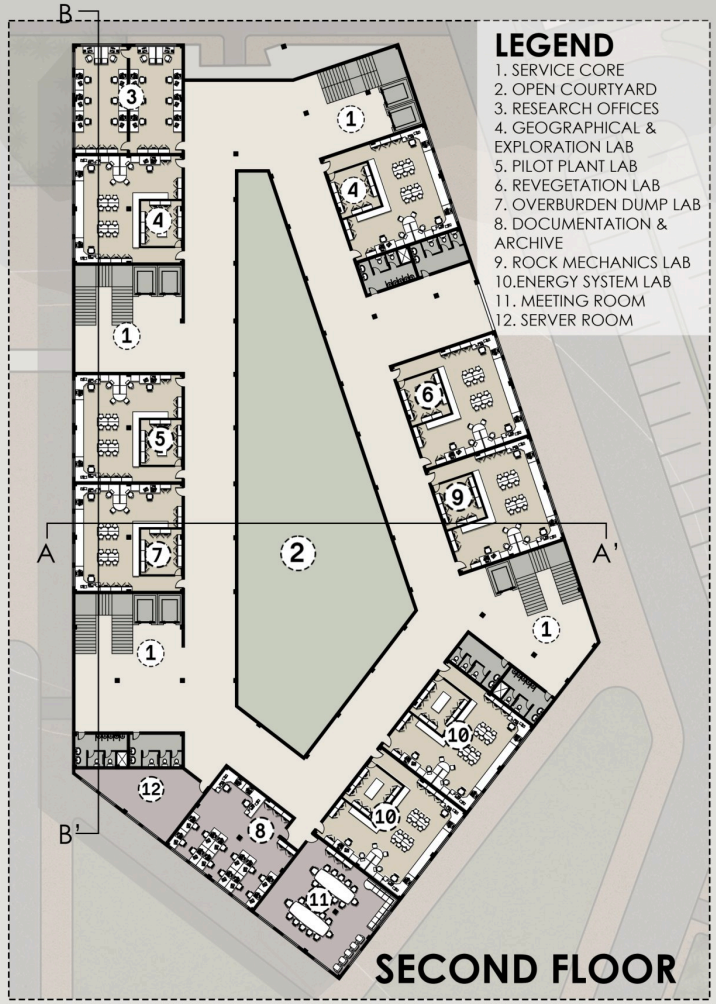
GROUND FLOOR



LEGEND

1. SERVICE CORE
2. OPEN COURTYARD
3. RESEARCH OFFICES
4. GEOGRAPHICAL & EXPLORATION LAB
5. PILOT PLANT LAB
6. REVEGETATION LAB
7. OVERBURDEN DUMP LAB
8. DOCUMENTATION & ARCHIVE
9. ROCK MECHANICS LAB
10. ENERGY SYSTEM LAB
11. MEETING ROOM
12. SERVER ROOM

FIRST FLOOR



LEGEND

1. SERVICE CORE
2. OPEN COURTYARD
3. RESEARCH OFFICES
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5. PILOT PLANT LAB
6. REVEGETATION LAB
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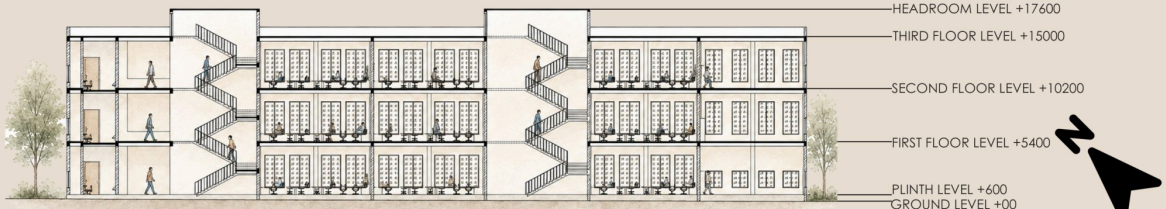
8. DOCUMENTATION & ARCHIVE
9. ROCK MECHANICS LAB
10. ENERGY SYSTEM LAB
11. MEETING ROOM
12. SERVER ROOM

SECOND FLOOR

BUILDING SECTIONS

SECTION AA'



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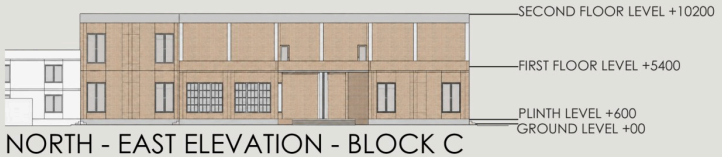
ELEVATIONS SCALE - 1:250

SOUTH - WEST ELEVATION - BLOCK C&D

NORTH - EAST ELEVATION - BLOCK A&B



NORTH - EAST ELEVATION - BLOCK C

NORTH - EAST ELEVATION - ADMIN AND BLOCK D

NORTH EAST ELEVATION - EXHIBITION BLOCK

NORTH - EAST SITE ELEVATION

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